Multivariate Data Analysis - MESIO UPC

Introduction, Exploratory Data Analysis and Preprocessing

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BARCELONATECH

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Multivariate Data Analysis

Topics part I

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4. Principal Component Analysis
5. Factorial Analysis
6. Distances in Statistics
7. Correspondence Analysis
8. Multiple Correspondence Analysis
9. Canonical Correlation Analysis
10. Multidimensional Scaling

Recommended books

- Aluja Banet, Tomas y Morineau, Alain , *Aprender de los Datos: El Análsis de Componentes Principales*, EUB , 1999 , ISBN:84-8312-022-4.
- Johnson, Richard A.; Wichern, Dean W. , *Applied Multivariate Statistical Analysis*, Prentice Hall , 1998 , ISBN:0-13-834194-X.
- Manly, B. F. J., & Navarro Alberto, J. A. (2024). *Multivariate Statistical Methods: A Primer* (5th ed.). Chapman and Hall/CRC, ISBN: 9781032591971
- Peña, Daniel; *Análisis de datos multivariantes*, McGraw Hill , 2002 , ISBN:84-481-3610-1.
- Husson, François; Lê Sébastien and Pages, Jérôme. *Exploratory Multivariate Analysis by Example Using R*. CRC

Resources

- R
- R Studio
- Wikipedia Topic MVA
- FactoMineR
- R for Data Science

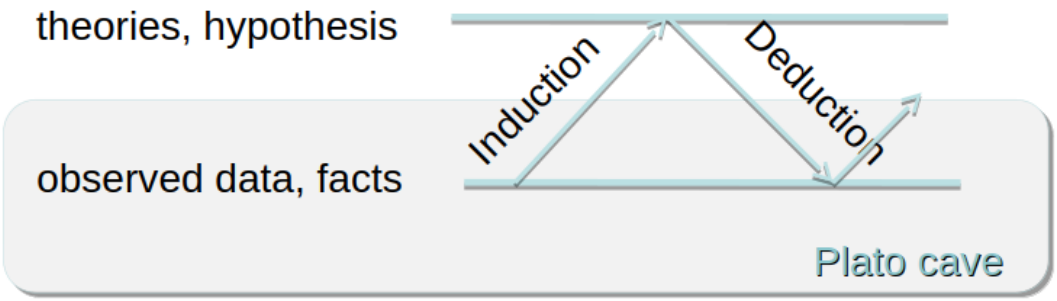
Grading

- Academic assessment will be based on:
  - Periodic exercises.
  - Exam for each part.
- Exercises will be assignments corresponding to the different units of teaching. They will be done in **pairs** and submitted as **reports**.
- Exams:
  - First part: 24 March
  - Second part: 3 June
    (The first part can be recovered on 3 June after the second part exam)
- Evaluation: 0.425 First exam + 0.425 Second Exam + 0,075 First part exercises + 0.075 Second part exercises

Introduction

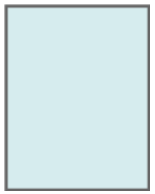
The need of data

Learning is an iteration process between the real world facts and the hypothesized world of theories.



MVA puts the focus on the inductive phase of learning (extract useful information from data).

Data = Fit + Noise

*Usually data is represented in flat files (matrices): 

Data (I)

Any stored data from any process always contains information about the generating phenomenon (**statistical regularity**).

Goal

To **extract useful information** (model, patterns, associations, trends, clusters, etc.) hidden in the data.

There is plenty of data stored in data bases... and will never be analyzed.

Now-a-days, massive volumes of data are generated and stored every minute; this 'data flood' (Big Data) has transformed data into a treasure for organizations.

Data (II)

Data analysis can optimize any process, yet it is often underutilized for simple reporting.

Storage the data in data bases and perform SQL queries for reporting is not enough but is an important step in the process.

The process of centralizing historical data from various transactional systems into a consistent format is known as **data warehousing**.

Trends that caused the data flood

- Exponential increase of data generation and storage.
  - Business transactions, scientific data, web, text, e-mail, e-commerce, social networks, etc.
- New formats:
  - Relational tables, non-structured data bases, text, image, video, sound, etc.
- Real time:
  - Streaming data, sensors, wereables, etc.

Trend: From economy of production to economy of online services.

Data (III)

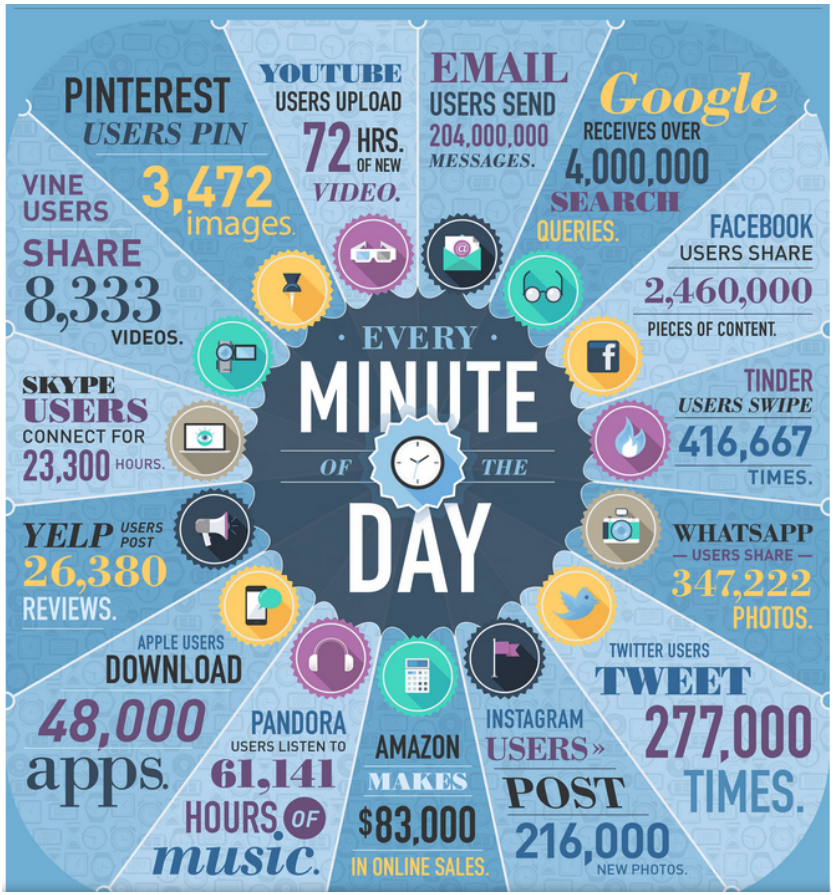
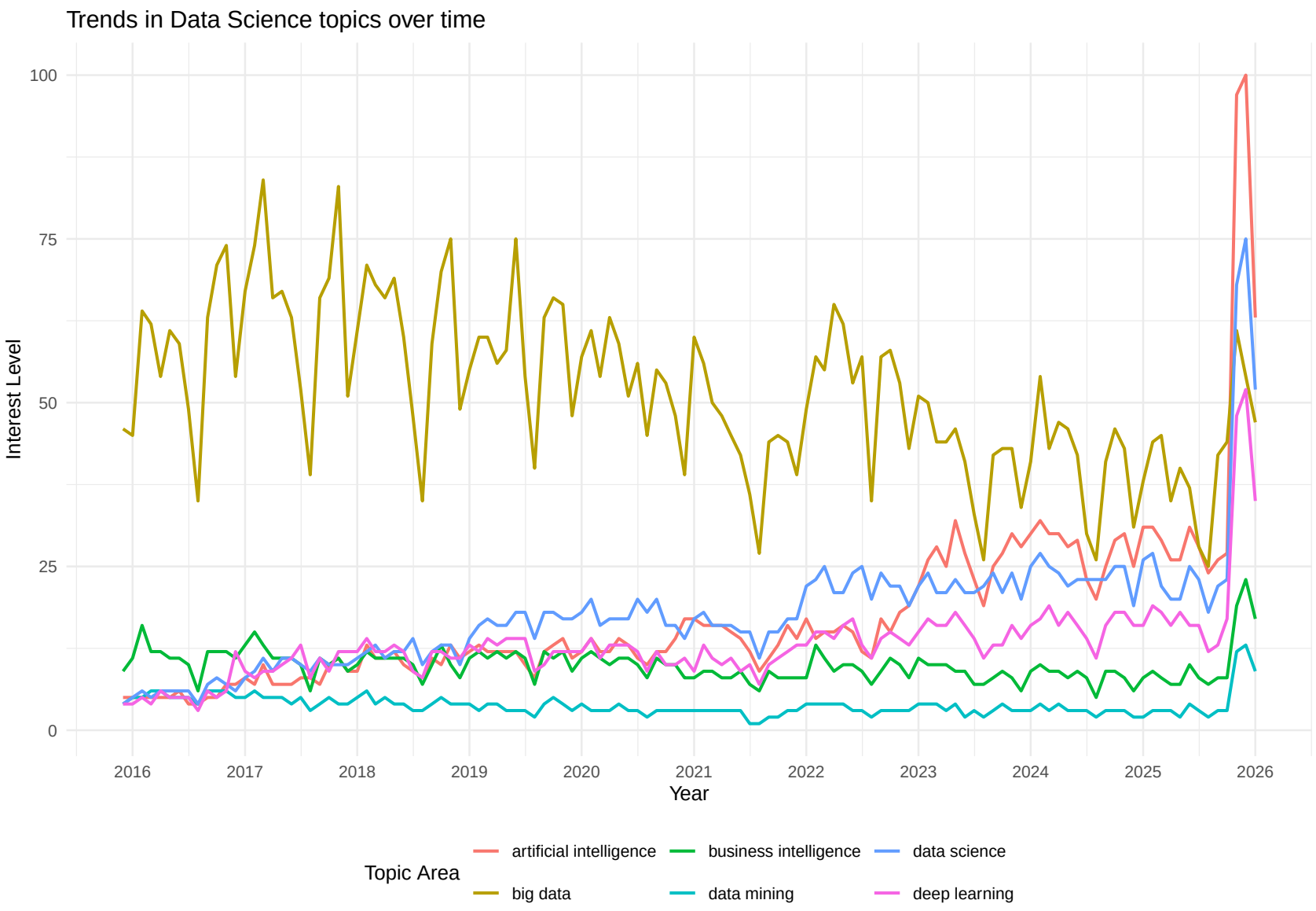


Figure 1: Image from domo: data never sleeps

Data (IV)


Trends in Data Science topics over time
Interest Level
100
75
50
25
0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Year
Topic Area
artificial intelligence
business intelligence
data science
big data
data mining
deep learning

Historical perspective

1. **Till the 70's**: Small data sets. Data collected according to an experimental design or a representative random sampling. Objective: To test a previously established hypotheses. Statistics time.
2. **From 70's**: Generalization of computers in companies. Installation of information systems (ERP's). Increasing trend of storing data. Machine Learning development. Development of the "french approach": "l'Analyse des données" school (Multivariate exploratory data analysis).
3. **From 90's**: Entrepreneurial interest in analyzing the large collected amounts of data. (CRM systems). Data Mining and Business Intelligence.
4. **From 2000**: First user-friendly data mining software (Clementine). Text mining, web mining, non structured data. R software.
5. **From 2010**: Big Data movement. Analysis of social networks, stream data, sensor data, image data, Deep Learning.
6. **From 2020**: Artificial Intelligence.

Statistics and Data Mining

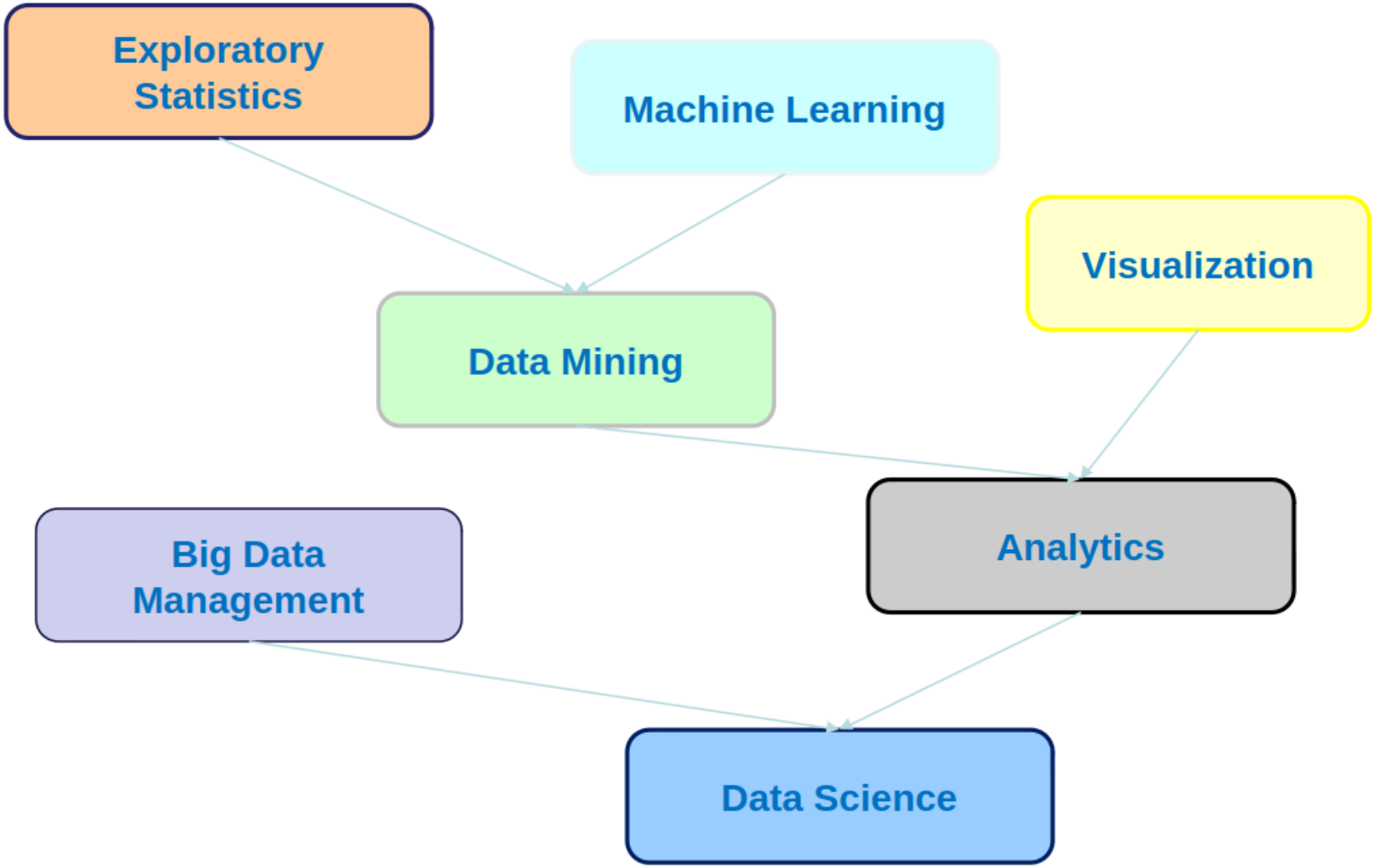
"*We are drowning in information but starved for knowledge*" - John Naisbitt, "Megatrends" (1982)

"*Statistics: Methodology for extracting information from data and expressing the amount of uncertainty in decisions we make*" - C.R. Rao, 1989

"*Data mining is the process of identifying valid, novel potentially useful and understandable patterns in data*" - Fayyad 1997, KDDB

"*Data Mining is the natural continuation of the exploratory approach of Statistics*" - Tukey (1962), Benzecry (1965)

Subjects around data



New roles in the industry

The abundance of data has led to new jobs in the industry:

- Data scientist.
- Data architect.
- Machine Learning Operations (MLOps) engineer.
- Prompt Engineer.
- AI Ethicist.

Interdisciplinary

Jobs created around data demand people with different skills:

- **Computer science**: Develop algorithms to manipulate data.
- **Statistics**: Explore and communicate data.
- **Machine Learning**:Use data to classify observations and provide predictions.
- **Domain knowledge**: Knowledge about the data at hand.

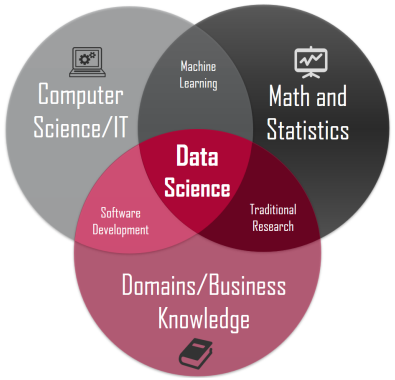


Figure 3: Image source

Roseta stone (from L. Lebart, 1995)

STATISTICS	MACHINE LEARNING
VARIABLES	ATTRIBUTES (DB: FIELDS)
INDIVIDUALS	INSTANCES (DB: REGISTERS)
EXPLANATORY VARIABLES, PREDICTORS	INPUT
RESPONSE VARIABLES	OUTPUT (TARGET), CONCEPT
MODEL	NETWORK, TREE, ...
COEFFICIENTS	WEIGHTS
FIT CRITERIUM (OLS, WLS, ML)	COST FUNCTION
ESTIMATION	LEARNING (TRAINING)
CLASSIFICATION ('CLUSTERING')	UNSUPERVISED CLASSIFICATION
DISCRIMINATION	SUPERVISED CLASSIFICATION

The data

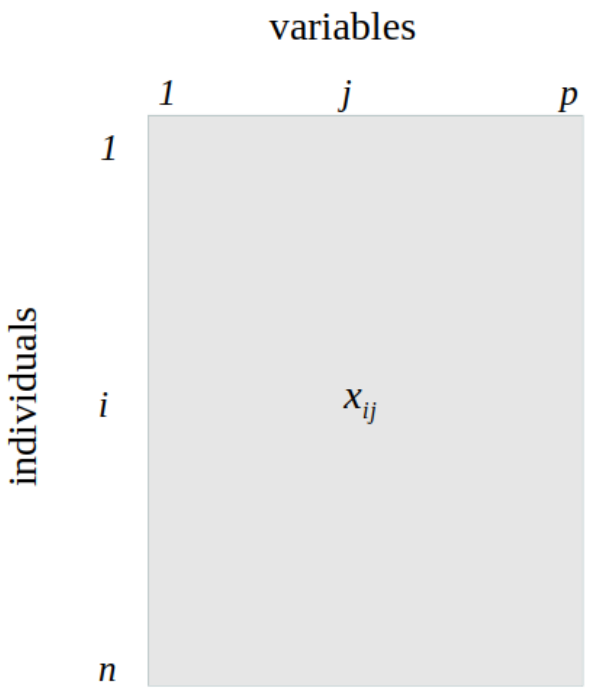
Multivariate data

Multivariate data: Refers to a dataset where multiple independent variables or measurements are recorded for each individual, object, or observation. It's key characteristics are:

- **Dimensionality:** Each observation is represented as a vector of p variables, often denoted as $\mathbf{X} = (X_1, X_2, \dots, X_p)$.
- **Interdependence:** The primary goal is to understand how these variables interact with one another.
- **Structure**: Data matrix where rows represent the **individuals** (observations) and columns represent the **variables** (attributes).

Data File

Data is multivariate:



Different sources with same representation: tables, transactions, graphs, text, streams, images, etc.

Data Table

	Quantitative & Qualitative Attributes (p Variables)			
ID	Variable_1	Variable_2	Variable_3	Category
Obs_1	23.5	102	0.5	A
Obs_2	19.2	98	0.4	B
Obs_3	25.1	115	0.8	A
Obs_4	22.8	108	0.7	C
Obs_5	20.4	101	0.5	B

Data matrix

A multivariate data matrix, $\mathbf{X} \in \mathbb{R}^{n \times p}$:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} 23.5 & 102 & 0.5 \\ 19.2 & 98 & 0.4 \\ 25.1 & 115 & 0.8 \\ 22.8 & 108 & 0.7 \\ 20.4 & 101 & 0.5 \end{pmatrix}$$

each element x_{ij} is the value of the jth variable for the ith observation.

Rows

- Represent individuals or instances.
- Alternative names:
  - Sample, example, record, etc.
- Theoretically we can repeat the measures (add new rows) until forming the population.
- Individuals are the "thing" under study and used to perform classification, learn associations or clusters.
- They are characterized by a predetermined set of attributes (variables).

Columns

- (Random) **Variables** represent measurable characteristics of the individuals which can take different values (following a random distribution).
- Different **types** of variables: Binary, nominal, ordinal, interval, ratio, etc.
- The same variables have been measured on the set of individuals.

Types of variables

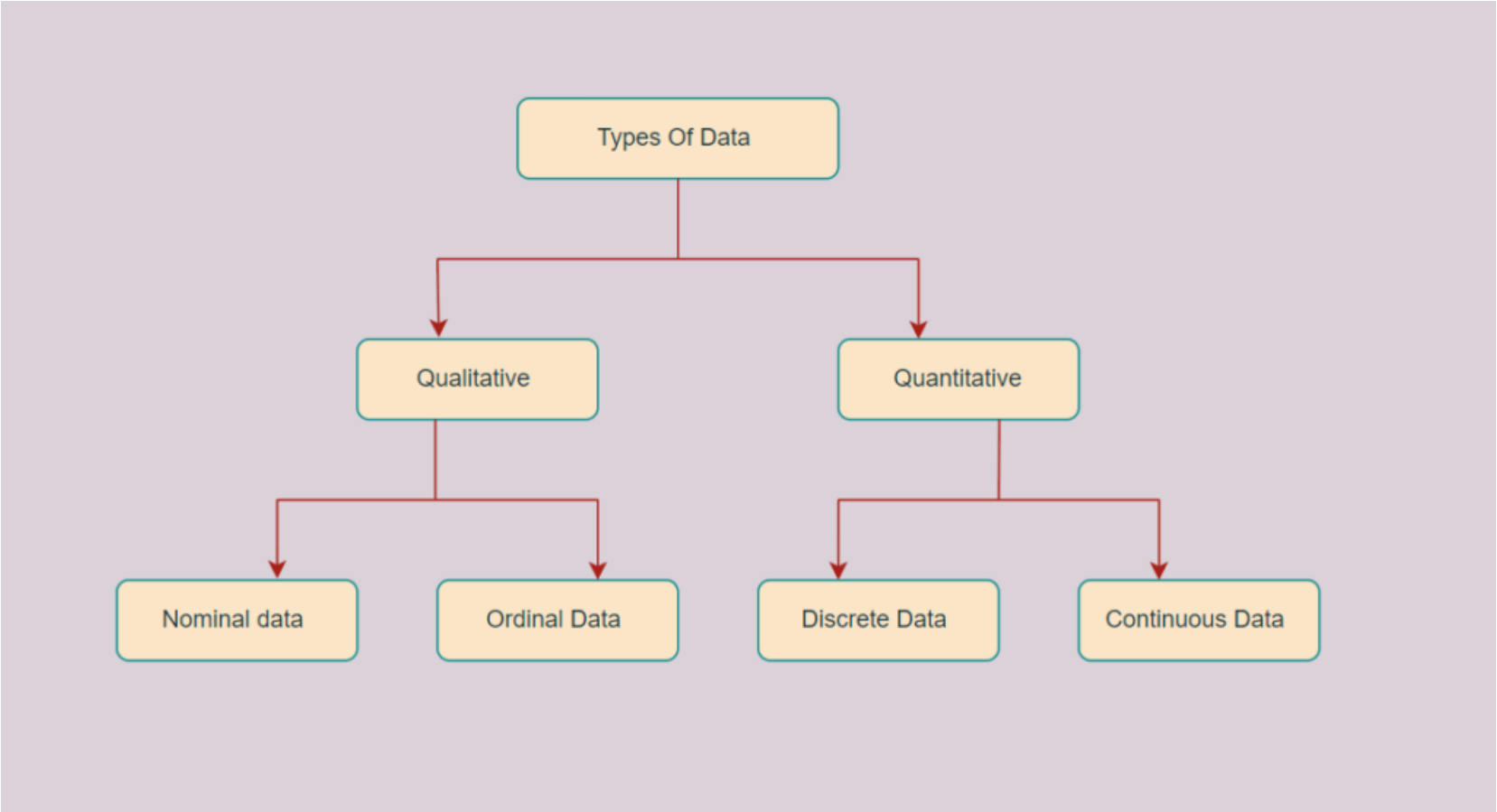


Figure 4: Image from https://www.mygreatlearning.com/blog/types-of-data/

Nominal variables (I)

- Values are distinct categories represented by **symbols** (labels or names).
- Example: “marital status”.
  - Values: “single”, “married”, “divorced” and “widow”.
- There is **no relation implied** between the possible values (no ordering or distance measure).
- Only **percentages** and **counts** can be calculated.
- Typical representation is in the form of **bar plots**.
- For numerical representation a **coding** is used:
  - One-hot encoding.
  - Dummy encoding.
  - Frequency encoding.
  - Target encoding.

Nominal variables (II)

Company	Cloud Provider (Nominal)	Region (Nominal)
1	AWS	North
2	Azure	South
3	Google Cloud	East
4	AWS	North
5	Azure	West

Nominal variables (III)

Cloud Provider	Frequency (n)	Percentage
AWS	2	40.0%
Azure	2	40.0%
Google Cloud	1	20.0%

Nominal variables (IV)

Ordinal variables (I)

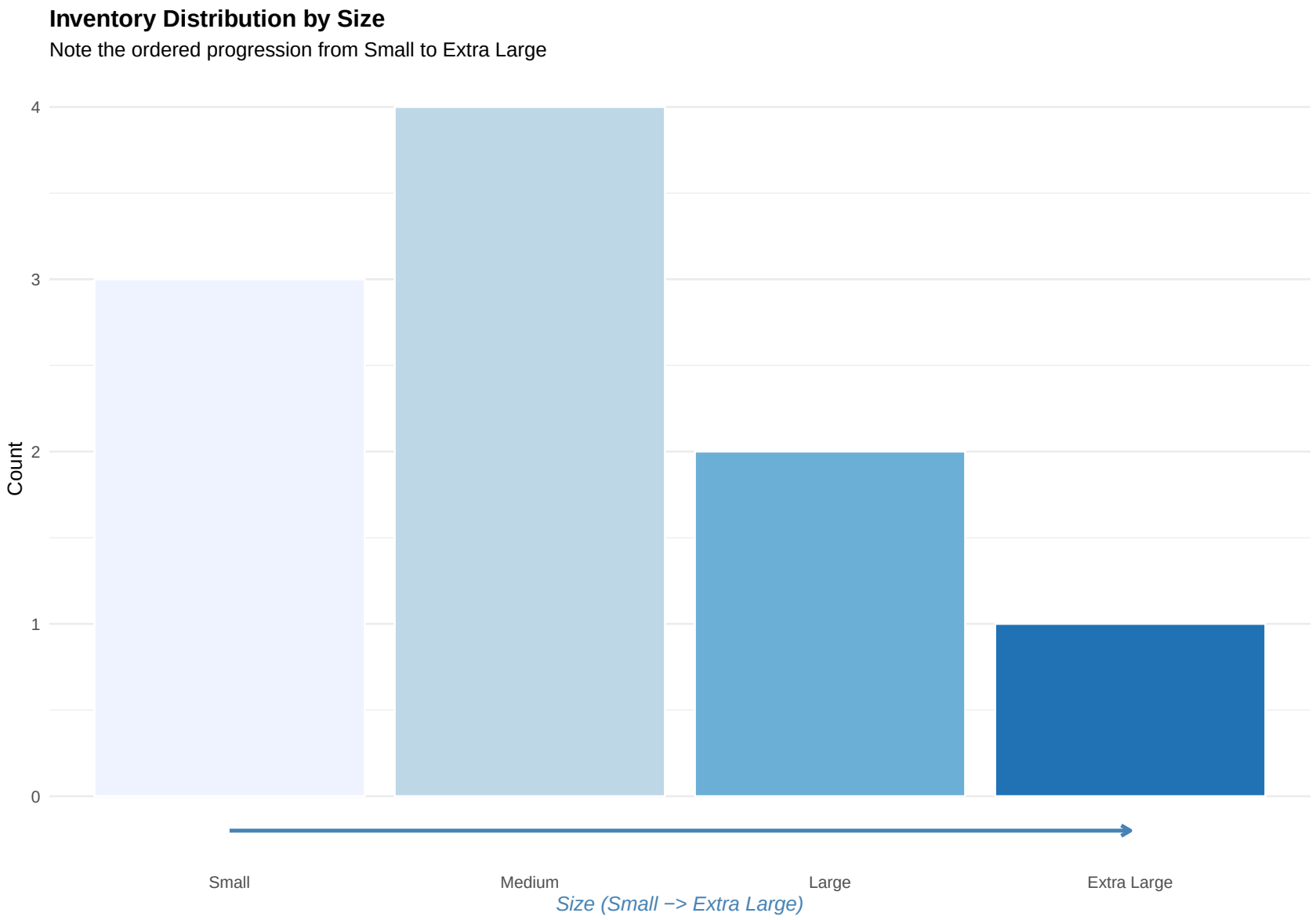
- Similar to nominal variables but there is an implicit order between the values, but no distance.
- Example: Clothes "size":
  - Values: "XXL" > "XL" > "L" > "M" > "S"
- Arithmetic calculations are not possible.
- Ordinal encoding: Assign an integer value in ascending ordering that represents order in the categories.
- Visualization using frequencies, percentages and bar plots and putting emphasis in the order.

Ordinal Scale: Shirt Sizes

Categories with a meaningful sequence

Size Category	Frequency	Share (%)
Small	3	30.0
Medium	4	40.0
Large	2	20.0
Extra Large	1	10.0

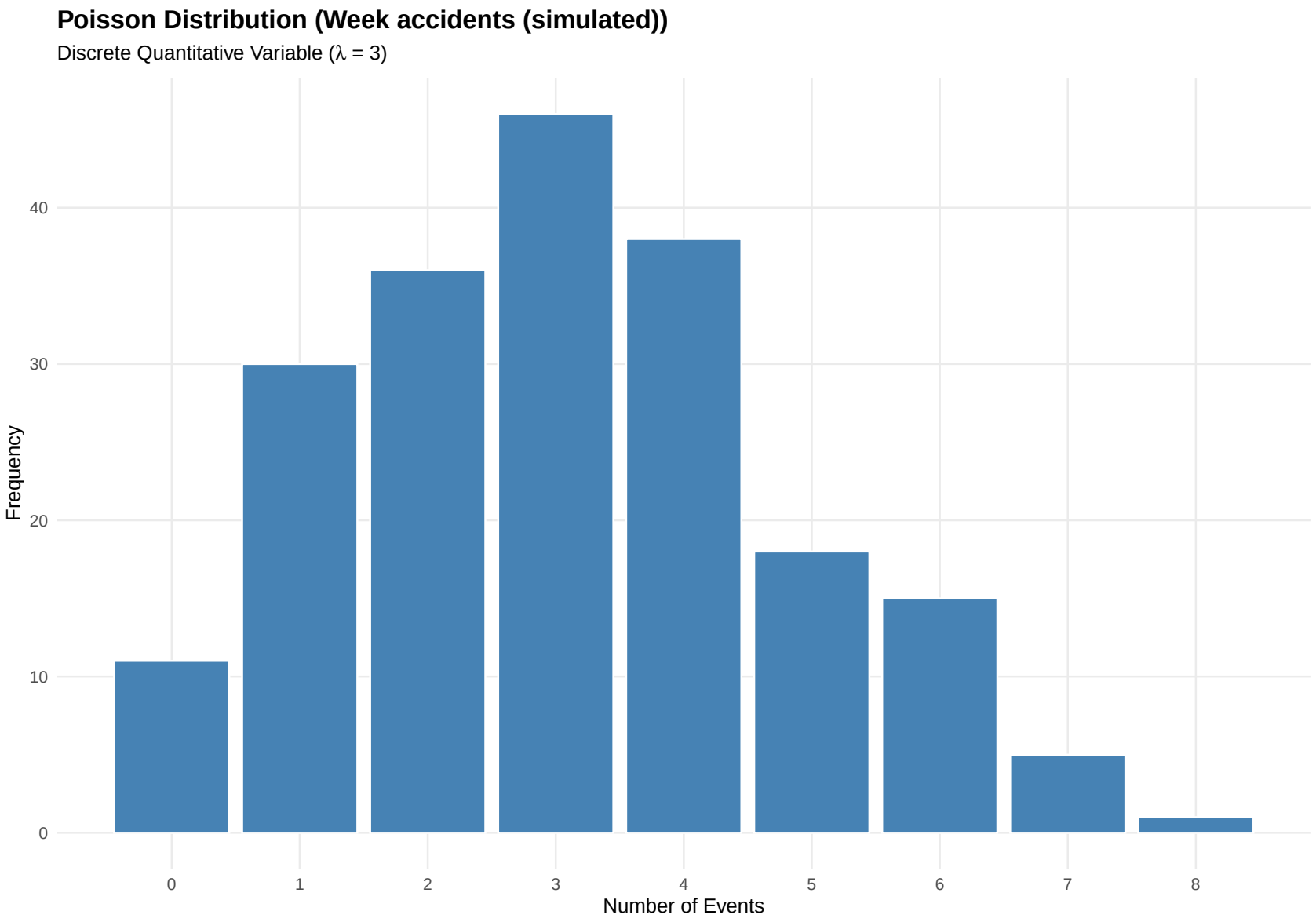
Ordinal variables (III)



Discrete variables (I)

- When the variable is a result of a count.
- Examples:
  - Number of words in a sentence.
  - Number of accidents in a week.
  - Number of products bought.
- Usually related to Poisson distributed random variables.
- Means and standard deviations are useful.

Discrete variables (II)



Continuous variables

- When the measured characteristic can take any value in the real line.
- Examples:
  - Weight.
  - Height.
  - Temperature.
- Different possible distributions but the most common one is the normal distribution.
- Common descriptors: means, median, range, IQR, variance, standard deviation, quarters, etc.

Variables as R types

- **Categorical:** Factor in R. The categories are called levels.
  - Binary.
  - Nominal.
  - Ordinal.
- **Continuous or discrete**: Numeric in R.
  - Count data (discrete).
  - Interval.
  - Ratio data.
  - Binary data.
  - Ordinal.

Be aware:

- Distinctions are often blurred (ordinal as discrete).
- Methods may only admit one type of variables.

Understanding the data

- Data is generated by an unobservable process. The mathematical abstraction is called **random experiment**.
- In its context, data has meaning and its related to the individuals and the variables.
- Variables meanings can be found in the **metadata**.
- Sometimes variables have different roles:
  - **Response** (target) variable: Variable that depends on others. Usually the outcome or the one to be explained. Denoted Y.
  - **Explanatory** (input) variable: Variable(s) used to explain the response variable. Denoted X_i for $i \in 1, 2, \ldots, p$

Warning

Disclaimer: The response variable is **context-dependent** and not always present. A variable's role is defined by the specific goals of the research.

Types of analysis

The goal of a multivariate analysis can change based on having a Response variable or not, its type and the research question at hand.

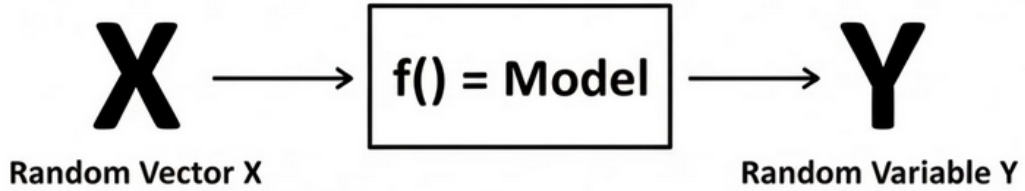
- **Descriptive analytics:** Methods to understand the data, find associations, exploration.
  - Profiling.
  - Clustering.
  - Visualization.
  - ...
- **Predictive analytics**: Methods to do predictions or classification (modeling).
  - Multiple regression.
  - Generalized Linear Models.
  - Discriminant Analysis.
  - Neural Networks.
  - ...

Types of modeling

Given a set of data from a random vector, $\mathbf{X} = (X_1, X_2, \dots, X_p)$, the goal is to find a function f such that $Y = f(\mathbf{X})$.

There are different ways to estimate f, namely methods (or learning algorithms in the Machine Learning literature), and which one is best given a data set depends on the data and on the research question. But also depends on the type of Y:

- **Categorical**: We will need to use a classifier method.
- **Numerical:** We will need to use a regression method.



The multivariate data analysis pipeline

1. **Preprocessing**: First look at the data. Cleaning (missing values? outliers? need of new features or more data? etc).
2. **Summary**: Univariate description and bivariate description.
3. **Multivariate exploration**: Visualization, clustering and profiling.
4. **Modeling**: Propose models and find the optimal one.
5. **Deployment**: Put the model into production.

Preprocessing

Data cleaning

The process of identifying and correcting errors, inconsistencies and inaccuracies in a dataset to ensure it is "fit for purpose". The goal is to improve data quality so that analysis and models remain reliable.

- **Handle missing values**: Delete records or use an imputation technique to fill them.
- **Remove duplicates**: Identify and merge redundant entries.
- **Standardization**: Ensure consistent format through the data.
- **Outlier detection**: Investigate extreme or anomalous values that may be errors.
- **Structural correction**: Fix typos, mislabeled classes, or handle incorrect data types.

Preparing the data for the analysis

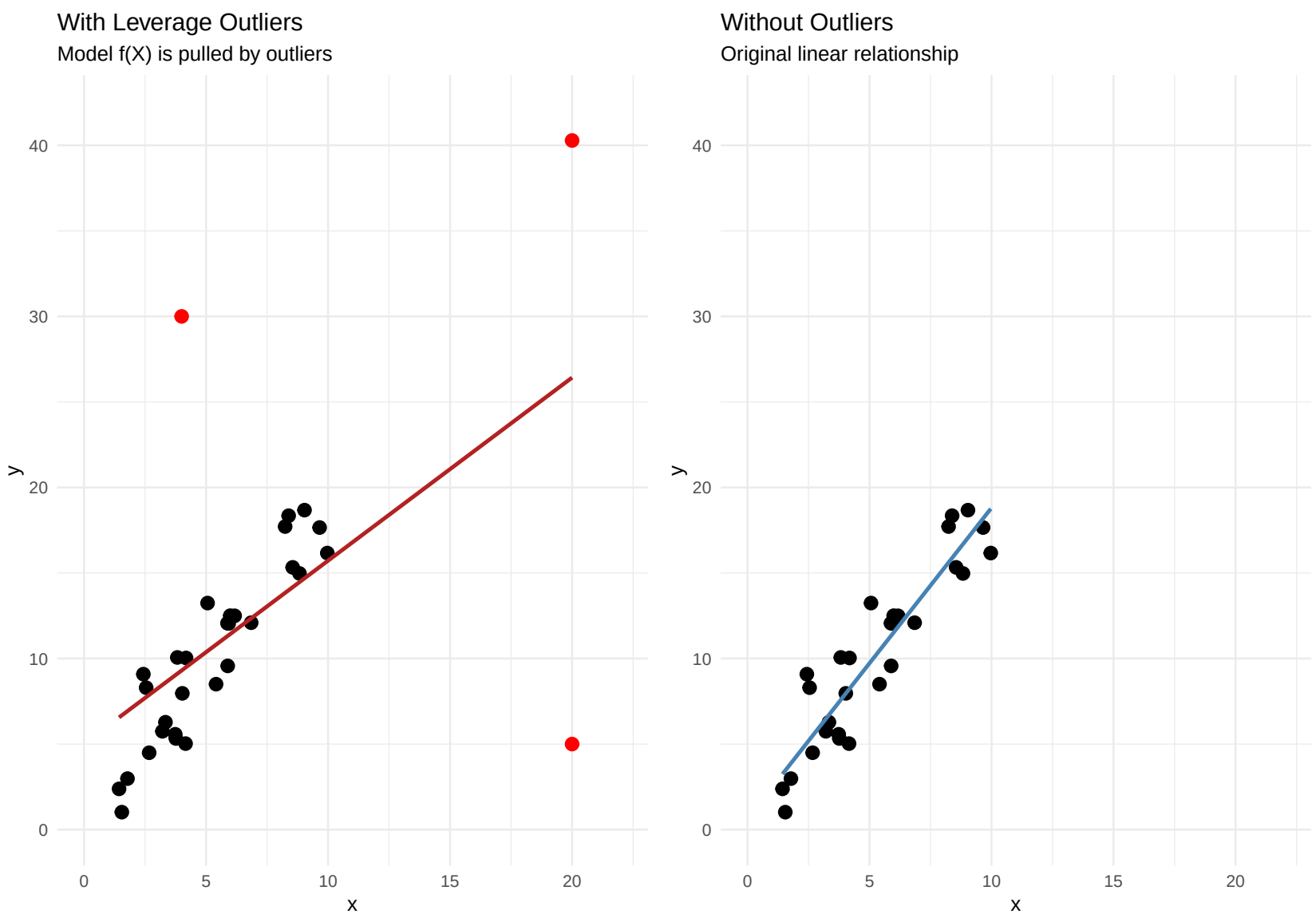
- **Feature selection**: Filtering the uninteresting variables.
  - Manual selection.
  - Automated selection.
- **Feature extraction**: Transforming or combining existing variables (often guided by domain expertise) to reduce dimensionality and highlight relevant patterns.
  - Manual engineering.
  - Automated extraction.
- **Transformations**: Transform the variables properly for the analysis:
  - Codification (categoricorical $\rightarrow$ numeric).
  - Recoding (numeric $\rightarrow$ categorical).
  - Normalization:

$$z_i = \frac{x_i - \bar{x}}{s_x}, \qquad \frac{x - min(x)}{max(x)}, \qquad log(x), \qquad ...$$

Data cleaning

- **Errors:** Logical errors and typos. Detect and correct.
- **Outliers:** Values that deviate from the variable expected range of values.
  - They can have a negative effect on the analysis (bias on the results).
  - They can be an error or not.
  - Approach: Do nothing, treat as missing data, talk with an expert, etc.
- **Missing values**: Cells without value. Several possible sources:
  - Non-response in sample surveys.
  - Dropouts in longitudinal data.
  - Approach: Different ways based on their origin.

Outliers (I)


With Leverage Outliers
Model f(X) is pulled by outliers
Without Outliers
Original linear relationship
y
x
0
10
20
30
40
0
5
10
15
20

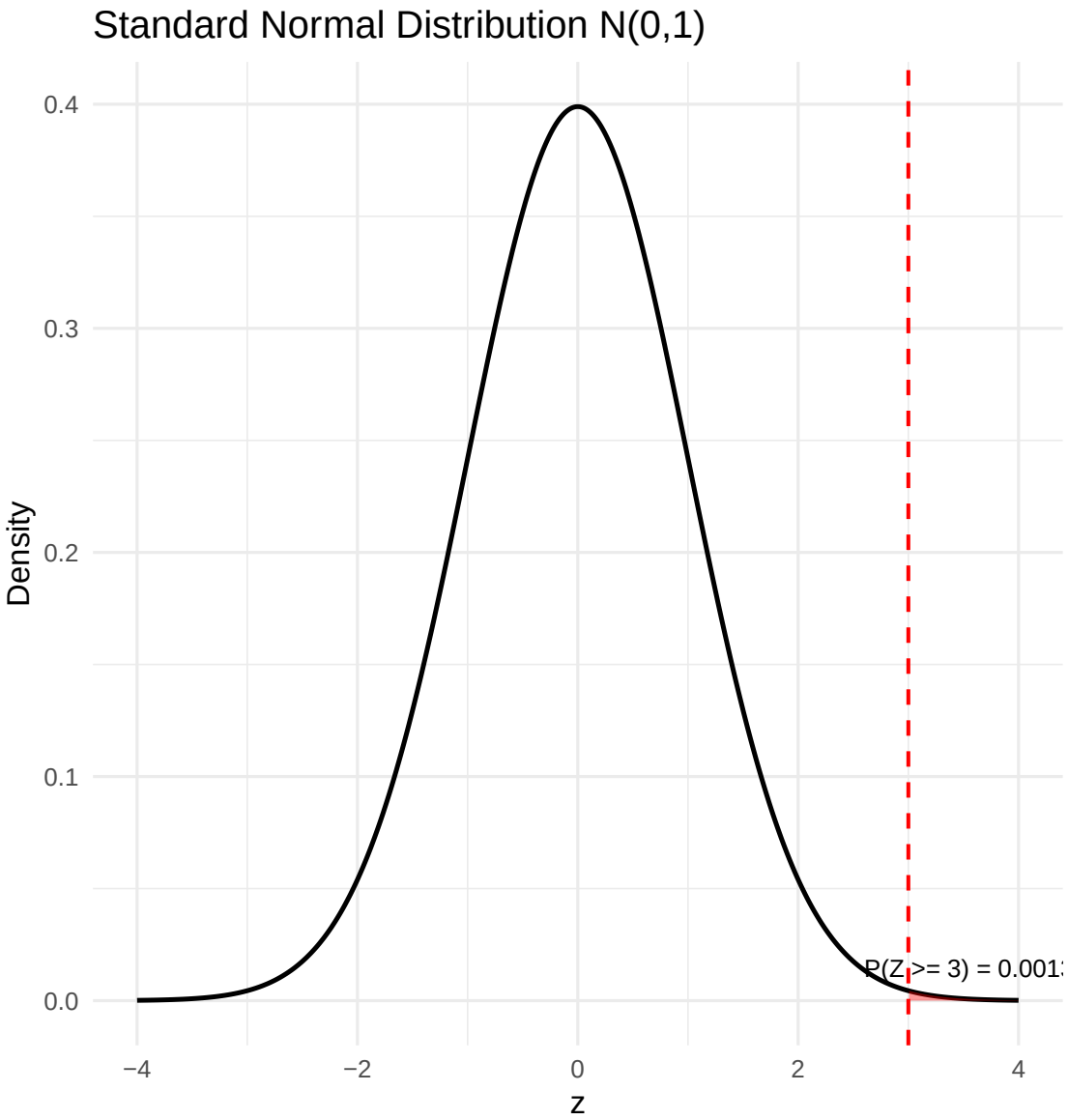
Outliers (II)

"An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism" - Douglas Hawkins.

Statistic intuition: Data is generated by an unknown process. The mathematical representations of these unknown processes are called probability distributions (e.g.: normal data follows a "normal generating data mechanism"). Then, an outlier can be:

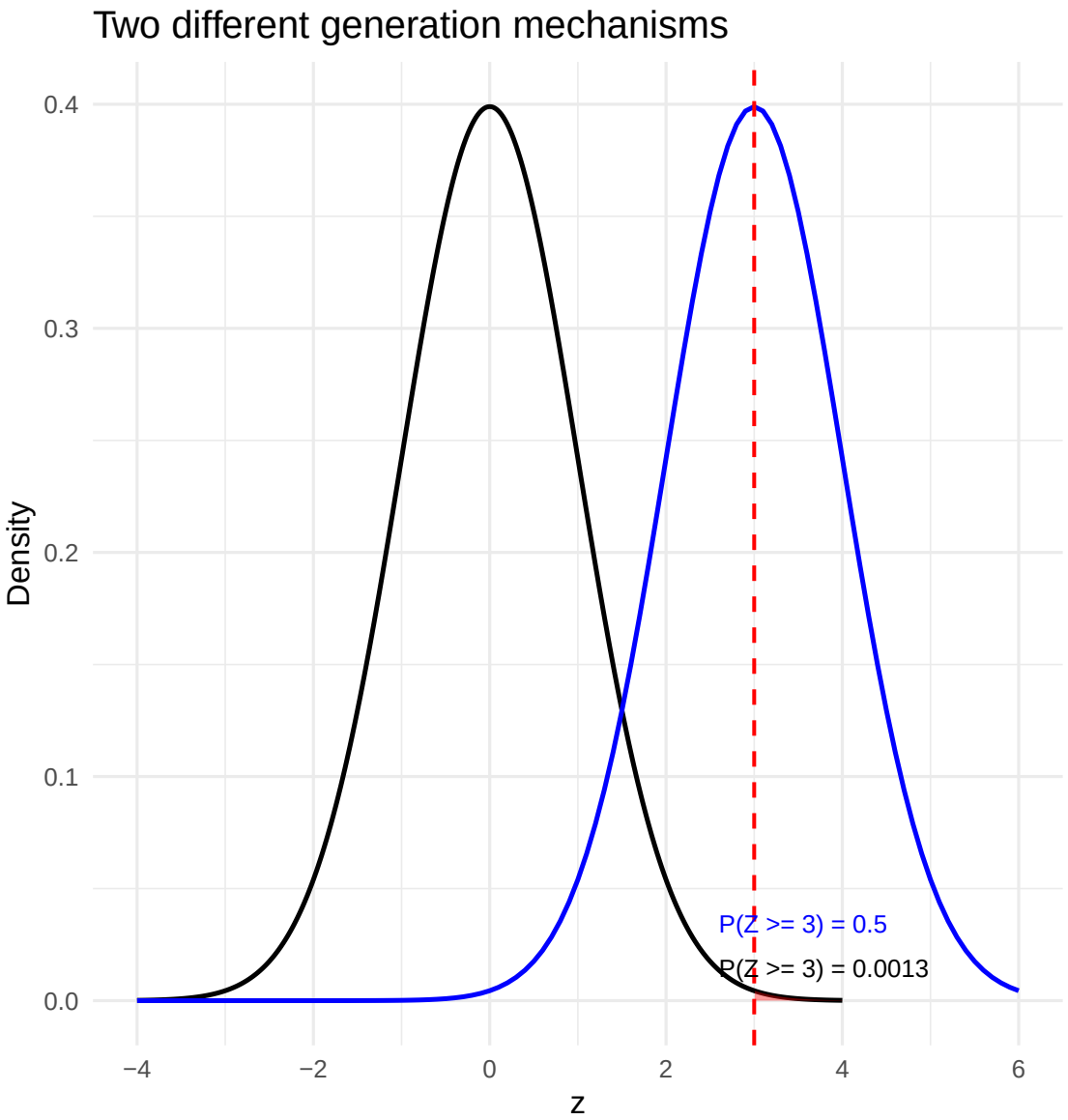
- Very **unlikely events** for the current generating mechanism (observations that sit on the tails of the distribution).
- Observations that have been **generated by a different mechanism**.
- Sampling **errors**.

Outliers (III)



z	P(Z<=z)	P(Z>=z)
−2	0.0228	0.9772
−1	0.1587	0.8413
0	0.5	0.5
1	0.8413	0.1587
2	0.9772	0.0228
3	0.9987	0.0013

Outliers (IV)



z	N(0,1)	N(3,1)
−2	0.0228	0
−1	0.1587	0
0	0.5	0.0013
1	0.8413	0.0228
2	0.9772	0.1587
3	0.9987	0.5

Univariate detection of outliers: Boxplot

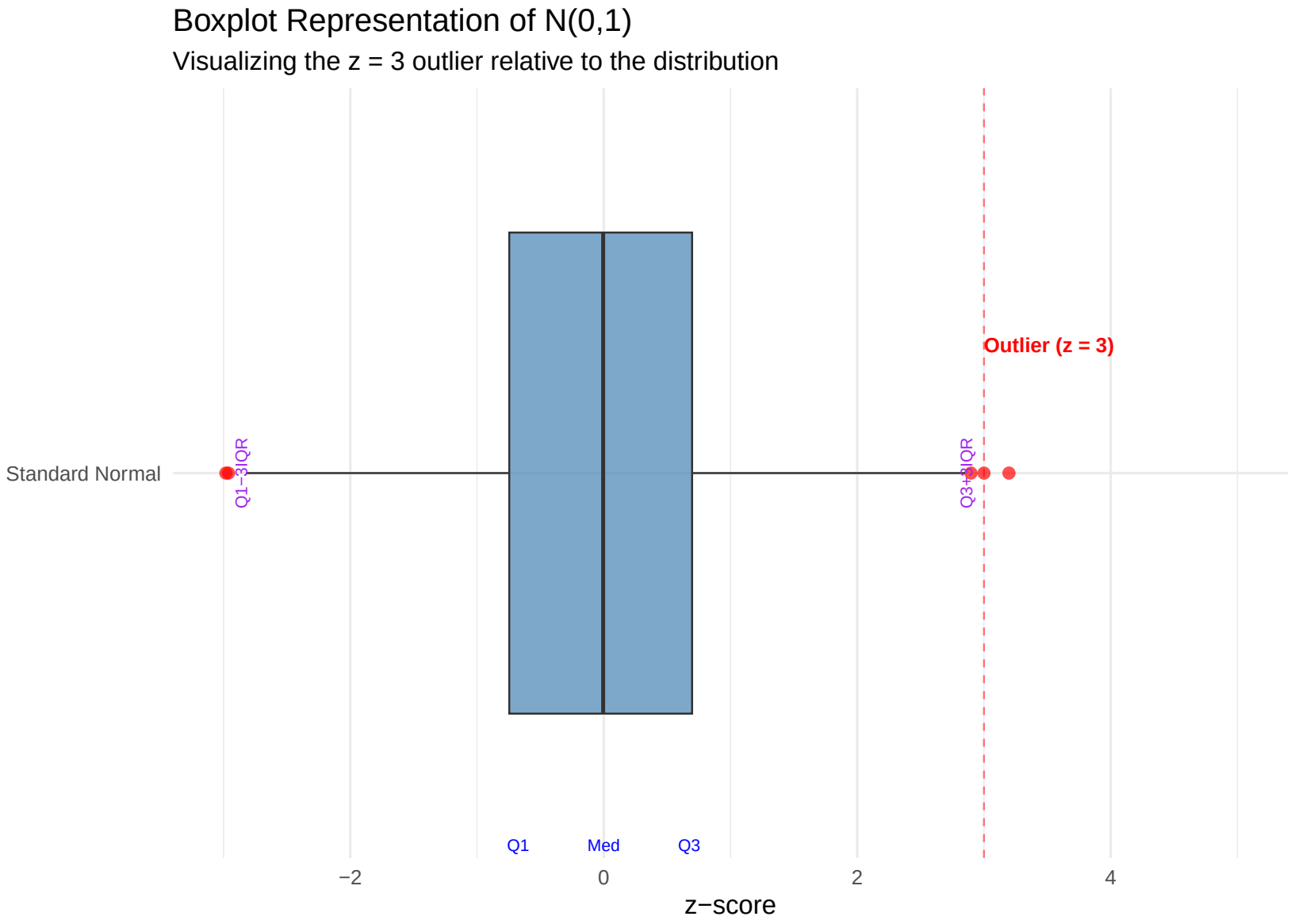
The Boxplot (Tukey, 1977) is a graphical display for exploratory data analysis, where the outliers appear tagged. Two types of outliers are distinguished: mild outliers and extreme outliers.

An observation x_i is declared an extreme outlier, if it lies outside of the interval $(Q1 - 3 \times IQR, Q3 + 3 \times IQR)$ where $IQR = Q3 - Q1$ is called the **Interquartile Range**. An observation x_i is declared a mild outlier if it lies outside of the interval $(Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR)$.

The numbers 1.5 and 3 are chosen by comparison with a normal distribution. If $X \sim N(\mu, \sigma)$ then:

- $Prob(X \geq Q_3 + 1.5IQR) = 0.003488302$ and
  $Prob(X \geq Q_3 + 3IQR) = 1.170971 \times 10^{-06}$

Boxplot example



Outliers in a multivariate context

1. When data is **multivariate** $\Rightarrow$ Univariate detection of outliers does not imply multivariate detection.
2. There is usually **more than one generating** mechanism/statistical process underlying the "normal" data. The basic model assumes only one "normal" generating mechanism where outliers are rare observations.
3. Outliers may represent a **different class** (generating mechanism) of objects, so there may be a large class of similar objects that are the outliers.

Multivariate detection of outliers

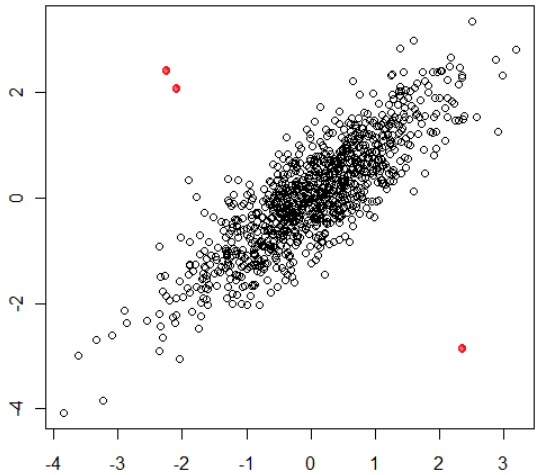


Figure 5: Outliers are multivariate. Univariate detection does not imply multivariate detection.

The detection of outliers is based on the computation of Mahalanobis distances relative to the data centroid, utilizing an iterative algorithm.

Mahalanobis distance

$$D_M^2(x, Q) = (x - \mu)^t \Sigma^{-1} (x - \mu)$$

A measure of the distance between a point x_i and a probability distribution Q. It is a **multi-dimensional generalization** of the idea of measuring how many standard deviations away is x from the center of gravity of a distribution Q.

If the generating process of the data is a multivariate Normal distribution then

$$D_M^2(x, Q) \sim \chi^2_{\tau=dim}$$

Under this circumstances, then one can define a threshold in order to mark observations as outliers, e.g.: $P(\chi^2_\tau \leq D_M^2) = 0.99$.

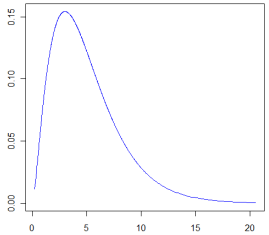


Figure 6: Chi square distribution.

Robust Mahalanobis distance

Problem: Σ and μ estimators are biased as they are influenced by the pressence of outliers.

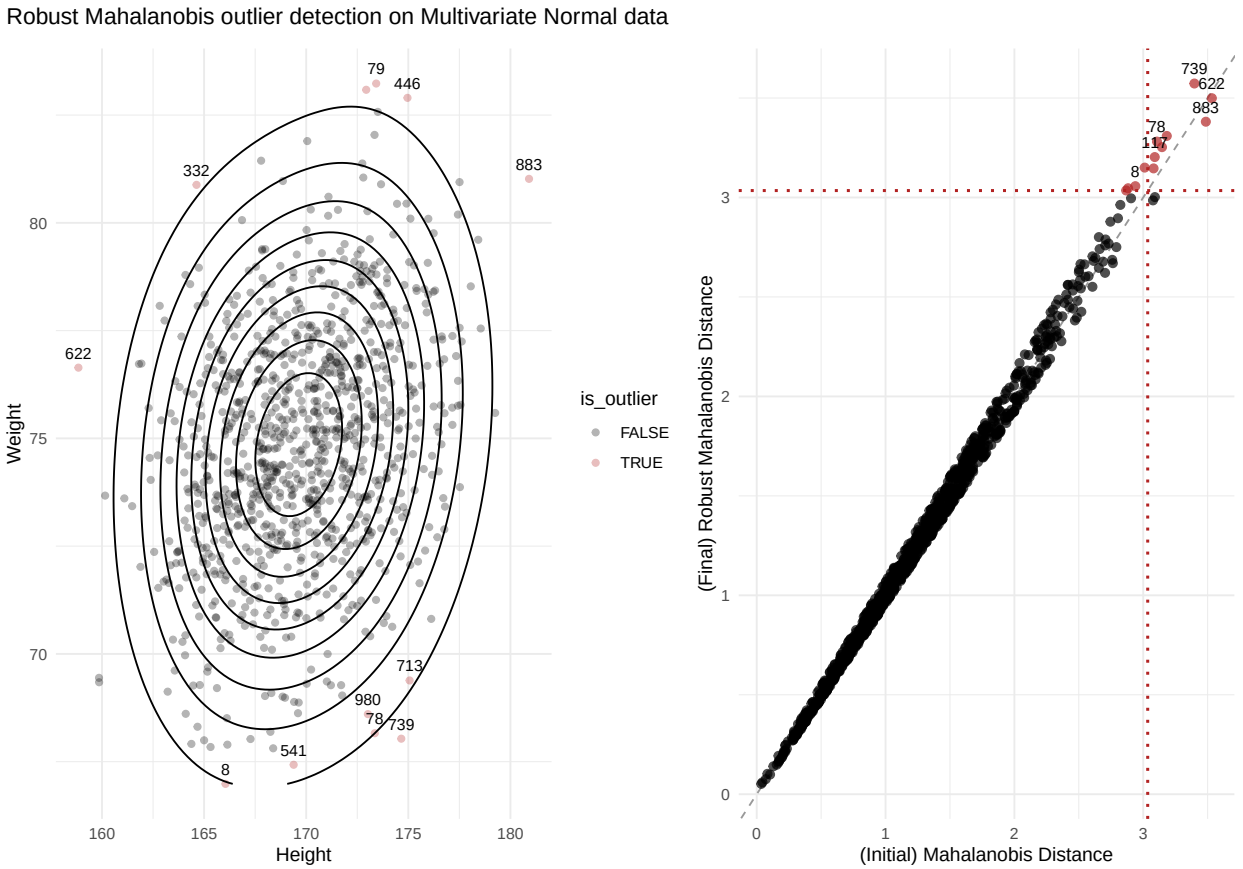
Solution: Use the Minimum Covariance Determinant (**MCD**) estimator.

Iterative algorithm:

1. Fix a value h as size of the data assumed not containing outliers. h must be larger than the number of variables. (Usually $0.75n$ is used or $\lfloor \frac{n+p+1}{2} \rfloor$, where n is the number of observations and p is the number of variables).
2. Take a subset H of size h.
3. μ and Σ are estimated using H.
4. Compute the Mahalanobis distances for all points.
5. Rank the distances and retain the h individuals with lower distance.
6. Update μ and Σ and iterate until convergence.
7. Repeat from point 2). Always keep the best MCD estimator found.
8. Using the Σ_{MCD} and μ_{MCD} compute the Mahalanobis distance to all the points and use a quantile threshold on the χ^2_τ distribution to mark the outliers.
9. Finally, the robust and classical Mahalanobis distances are compared via a scatterplot to identify potential outliers within the dataset.

Mahalanobis in R

R base has the function `mahalanobis` with arguments x (observation), $center$ (mean vector) and cov (covariance matrix) that computes the squared mahalanobis distance of the observation to the distribution. **Robust mahalanobis** is offered by different libraries like for example the `Moutlier`method in the `chemometrics`library.



Non parametric detection of outliers

Local Outlier Factor: LOF is an algorithm for identifying density-based local outliers (Breunig et al., 2000). It compares the local density of a point to the local densities of its neighbors and if it is significantly lower then it is marked as outlier. Calculated through four steps:

1. **k-distance and k-neighbor:** Identifying the distance to the k-th nearest neighbor.
2. **Reachability Distance:** A smoothing distance measure between two points.
3. **Local Reachability Density (LRD):** How close a point is to its neighbors (the inverse of the average reachability distance).
4. **LOF Score**: The ratio of the average LRD of the neighbors to the LRD of the point itself.

Local Outlier Factor

$$LOF_k(x_i) = \frac{\sum_{\text{nei}_{x_i}} \frac{\max d_k(x_i)}{\max d_k(\text{nei}^k_{x_i})}}{k}$$

where $d_k(x_i)$ is the reachability distance of x_i measured as $max\{k\text{-distance}(x_i), d(A, x_i)\}$ and k-distance is the k nearest neighbors of x_i.

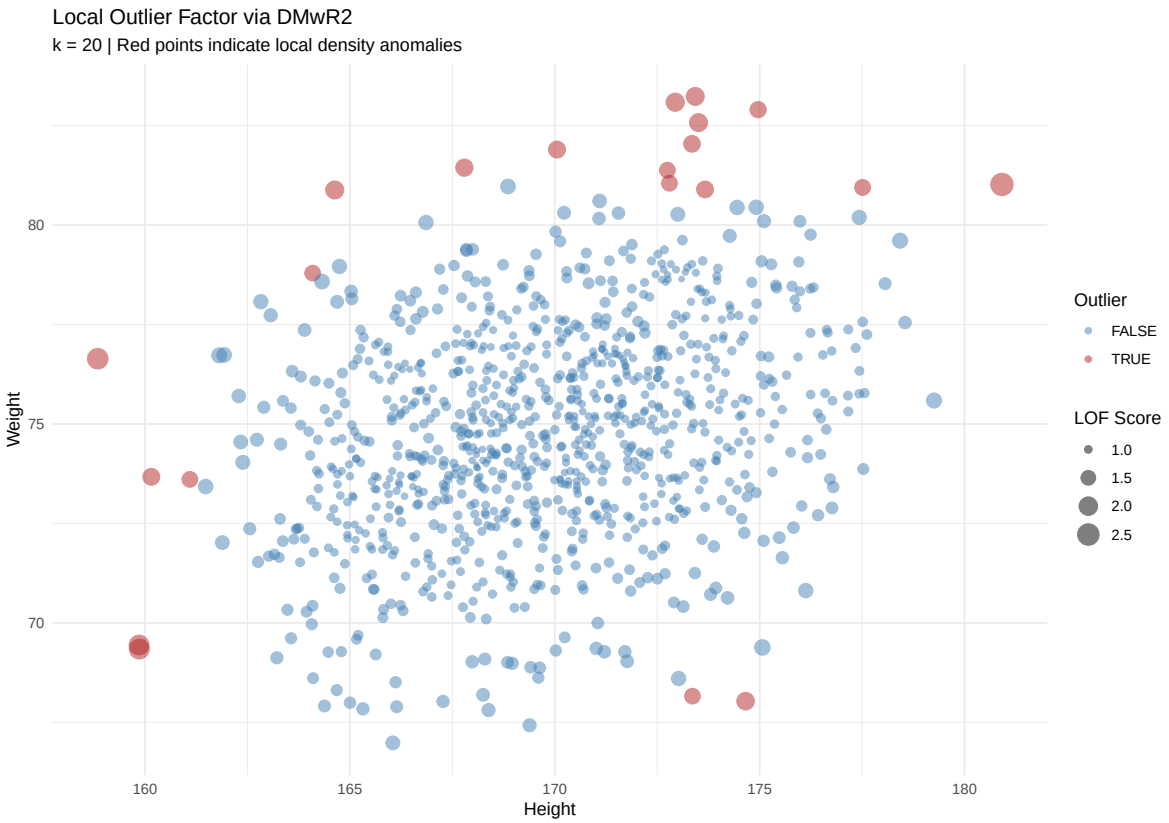
- $LOF \approx 1$: The point is similar to its neighbors.
- $LOF < 1$: The point is in a denser region than its neighbors.
- $LOF > 1$: The point is probably an outlier.

Warning

Note: While Mahalanobis has the whole dataset as reference, LOF acts locally.

Local Outlier Factor in R

In R we can use the method `lofactor` from the library `DMwR2`



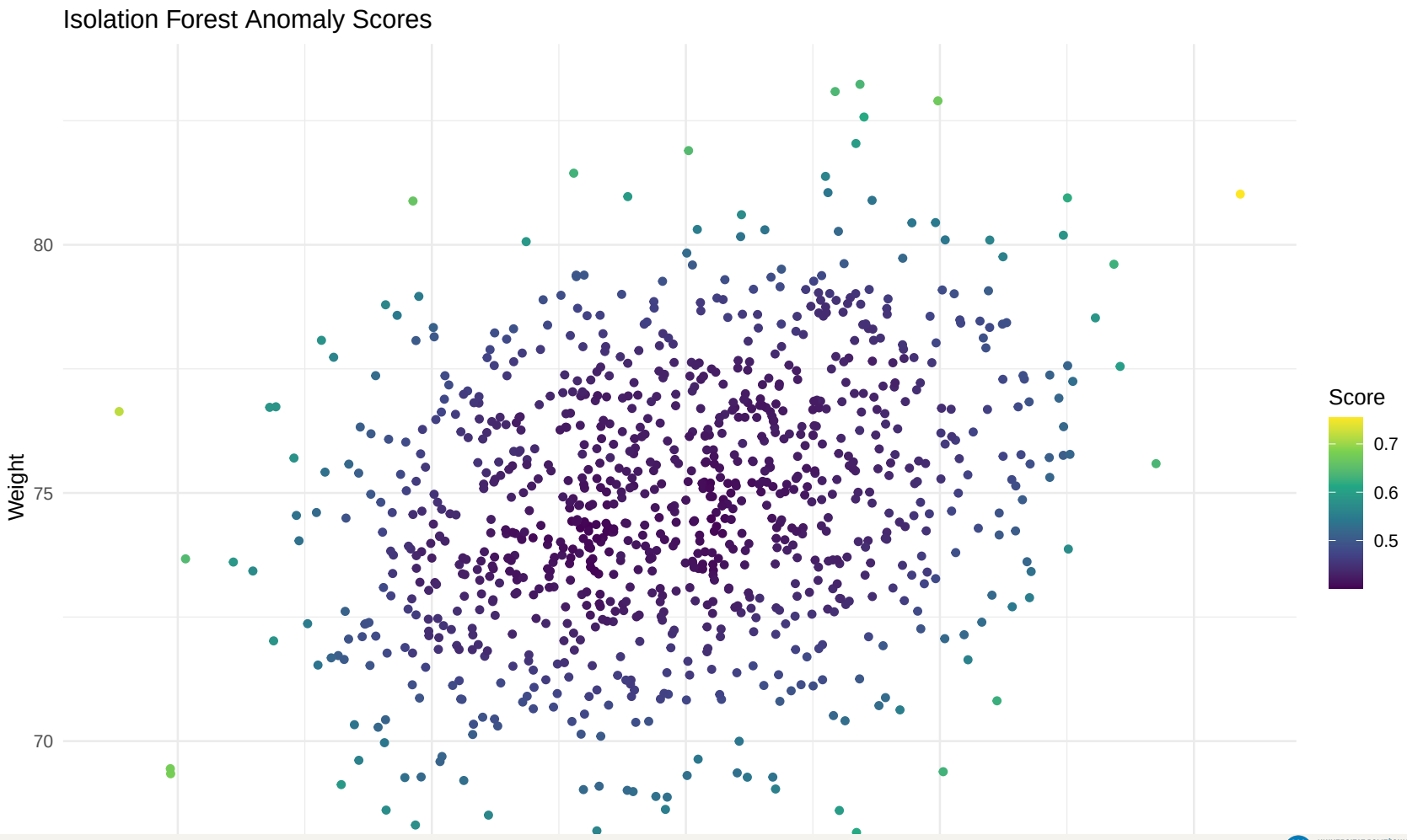
Outliers with Tree based methods

Isolation Forest: Unsupervised learning algorithm that detects anomalies by "isolating" observations. The core idea is that outliers are few and different, making them easier to isolate than normal points. It detects anomalies by randomly partitioning features. Observations with significantly shorter average path lengths are classified as outliers, as they require fewer recursive splits to be isolated from the rest of the distribution.

1. **Path Length:** $h(x)$.
2. **Average Path Length for n points:** $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$ where $H \approx ln(i)$ is the harmonic number and represents the expected path length for a normal point. Note: if $n \leq 1 \Rightarrow c(n) = 0$
3. **Anomaly score:** $s(x, n) = 2^{-\frac{\mathbb{E}[h(x)]}{c(n)}}$
   - 3.1 $s \approx 1$: The path length is very short. Mark the observation as outlier.
   - 3.2 $s < 0.5$: The path length is long. Likely a normal observation.
   - 3.3 $s \approx 0.5$: Indeterminate. Lot of samples with this value indicates that the dataset is clean or that it is not possible to discriminate randomness from data.

Isolation Forest in R

The `isotree` library in R has the `isolation.forest` method that performs the Isolation Forest algorithm. It needs the data and the number of trees to use.



Outliers detection with Principal Component Analysis

The Principal Component Analysis (PCA) is a fundamental technique in multivariate analysis. Its goal is to project high-dimensional data into a new coordinate system to simplify analysis. By reducing dimensionality and removing redundant information, PCA highlights the underlying structure of the data, allowing to identify observations that deviate from the general patterns.

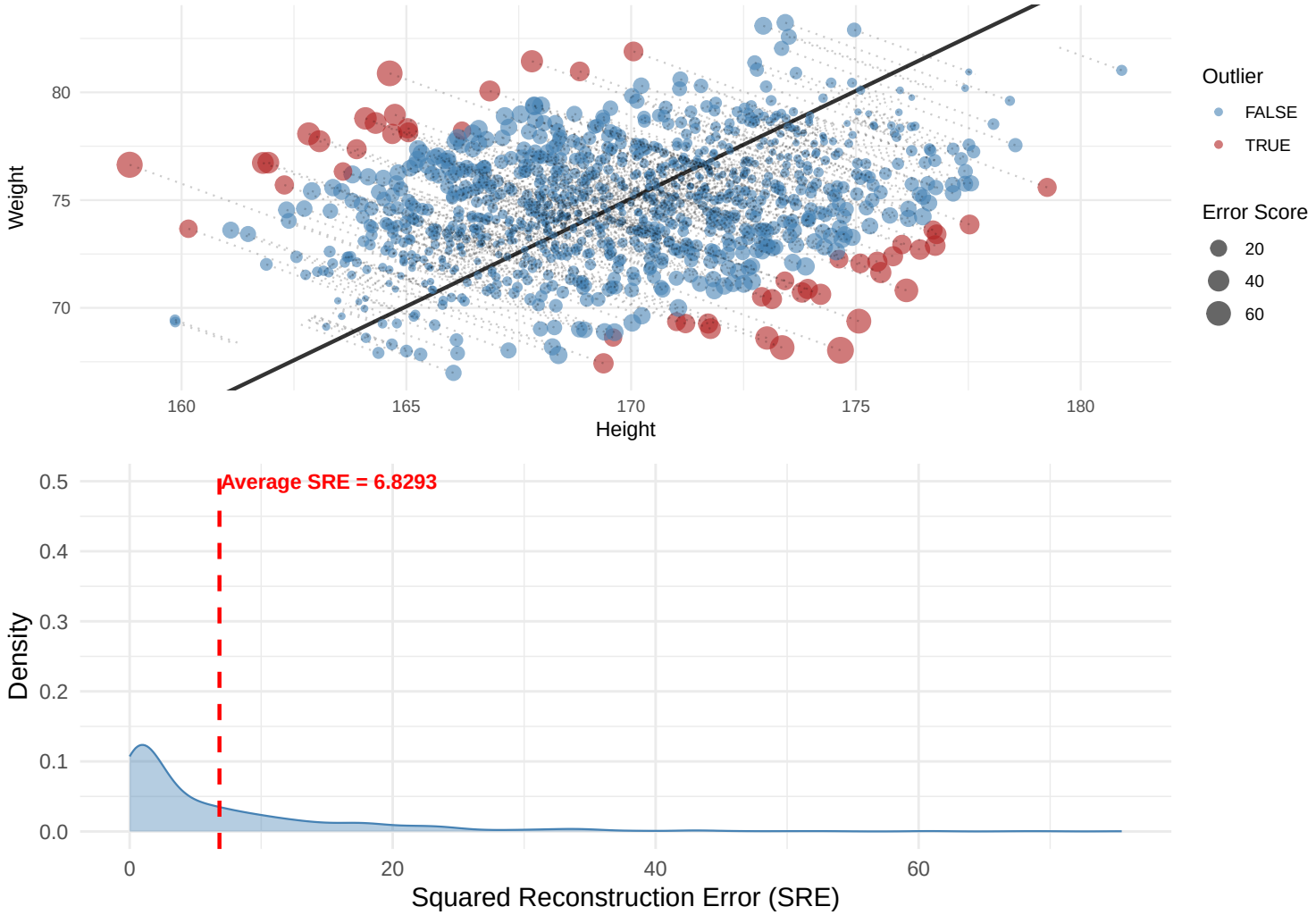
- PCA reconstruction formula:

$$\tilde{X} = V_{[,1:k]} D_{[1:k,1:k]} V^t_{[,1:k]}$$

- PCA squared reconstruction error for a point i: $SRE_i = \sum_j^p (x_{ij} - \tilde{x}_{ij})^2$
- Average Squared Reconstruction Error (ASRE) for the whole dataset: $\frac{1}{n} \sum_{i=1}^n SRE_i$

Outliers detection with PCA in R

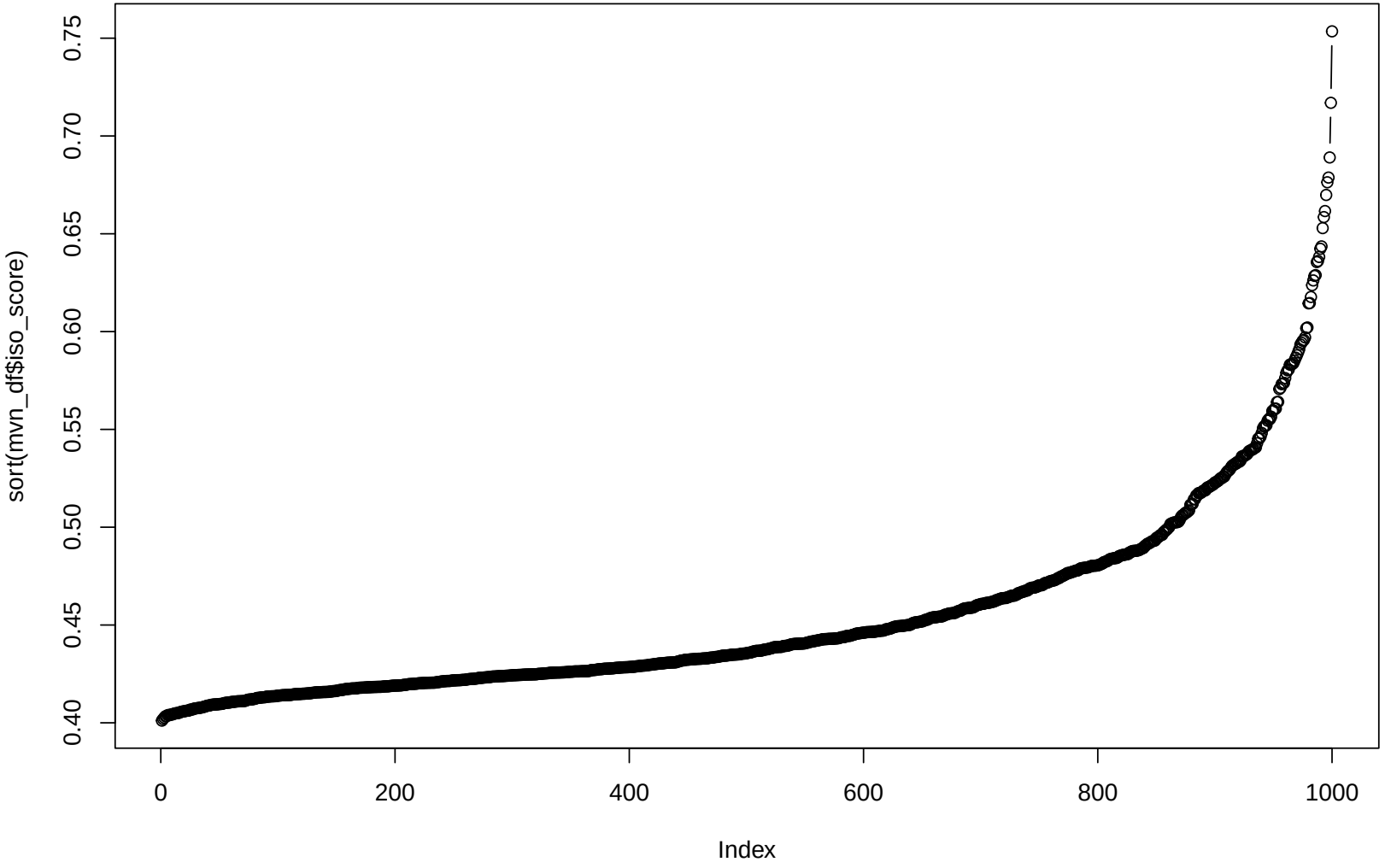
In R there are multiple ways to compute PCA from libraries or from base R. In base R we have the command `prcomp`.



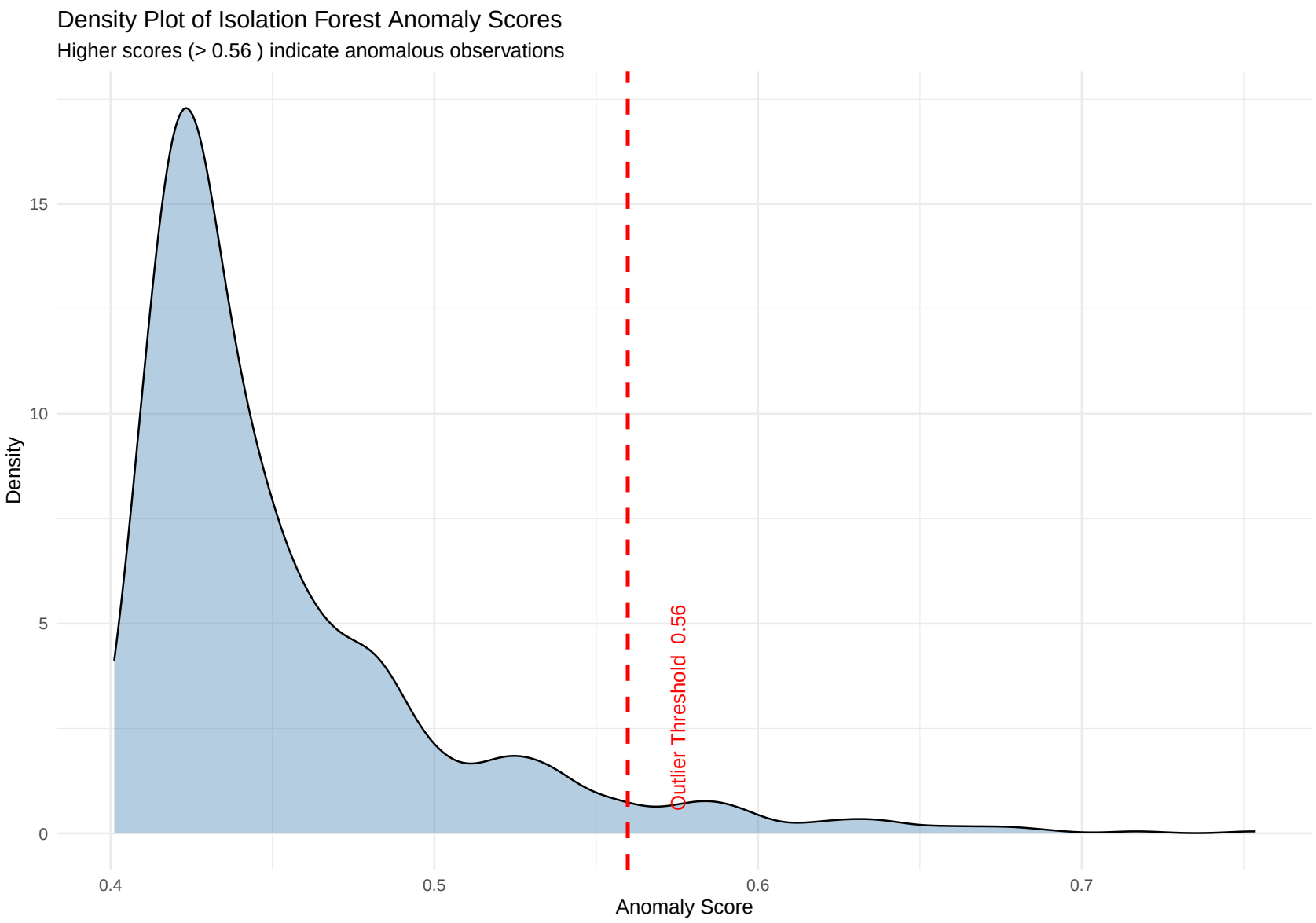
How many outliers?

- There will be always some observations that can be considered outliers.
- It is not worrying to see up to 1% of extreme outliers and up to 5% of mild outliers in any distribution.
- For methods like **Isolation Tree** or **Local Outlier Factor** we can set up the threshold by:
  - The **elbow method**: plot the anomaly scores from smallest to largest and put the threshold on the jump.
  - Looking for a extreme quantile in the anomaly score distribution, e.g.: 95%, 97.5% or 99%.

Elbow method



Quantile threshold



Outliers in practice

- In some analysis finding and understanding the outliers is the research question. Known as **Anomaly detection** in the literature.
- Outliers can bias the statistical methods.
- What to do with detected outliers?
  - Eliminate them (we lose data and we may not have enough expertise domain...). In some cases outliers are recursive.
  - Use weights so the importance of the observations in the methods is inversely proportional to its degree of being an outlier.
  - Mark outliers as "missing values" and treat them as missing data.
- Applications: Fraud detection, network intrusion, behavioral changes, etc.

Missing data

Missing data (or missing values) occurs when no data value is stored for a variable in an observation. They can appear as `NA`, `NaN`, `null` or with a placeholder like 999.

We must understand why there is missing data so we can take actions and fix it. Some causes:

- Data collection errors.
- Non-response in a survey.
- Attrition in longitudinal studies.
- Data entry mistakes.
- Structural missingness by design (inapplicable questions to some respondents).

! Important

Missing data is an indicator of **data quality**.

The missing data problem

Red: Missing values. Yellow: Drop-out. Green: Non response.

ID	Var_1	Var_2	Var_3	Var_4	Var_5
1	Value	Value	Value	Value	Value
2	Value	**NA**	Value	Value	Value
3	Value	Value	Value	Value	Value
4	Value	Value	Value	**NA**	Value
5	Value	Value	Value	Value	Value
6	Value	Value	**NA**	**NA**	**NA**
7	Value	Value	Value	Value	Value
8	**NA**	Value	Value	Value	Value
9	Value	Value	Value	Value	Value
10	**NA**	**NA**	**NA**	**NA**	**NA**

When missing data is a problem

1. Ignoring missing data can seriously **bias** the results.
2. Missing data represents a **loss of information** (waste of resources).
3. The impact of missing data depends on its **generating mechanism** (why some values are missing?)

It can be avoided with careful **planning** of data collection.

Steps to work with missing data

1. **Identify the missing data**:
   a. Assign a common code to the numerical variables.
   b. Assign a new category (missing) to a categorical variable.
2. **Check the quality of the information:**
   a. Count the number of missing per variable and rank them accordingly.
   b. The more the missing the less reliable is the information in the variable.
3. **Characterize the missingness mechanism:**
   a. Create a new variable counting the number of missings per individual.
   b. Profiling of global missingness analysis (individual) and compare both groups of cases (variables and individuals).

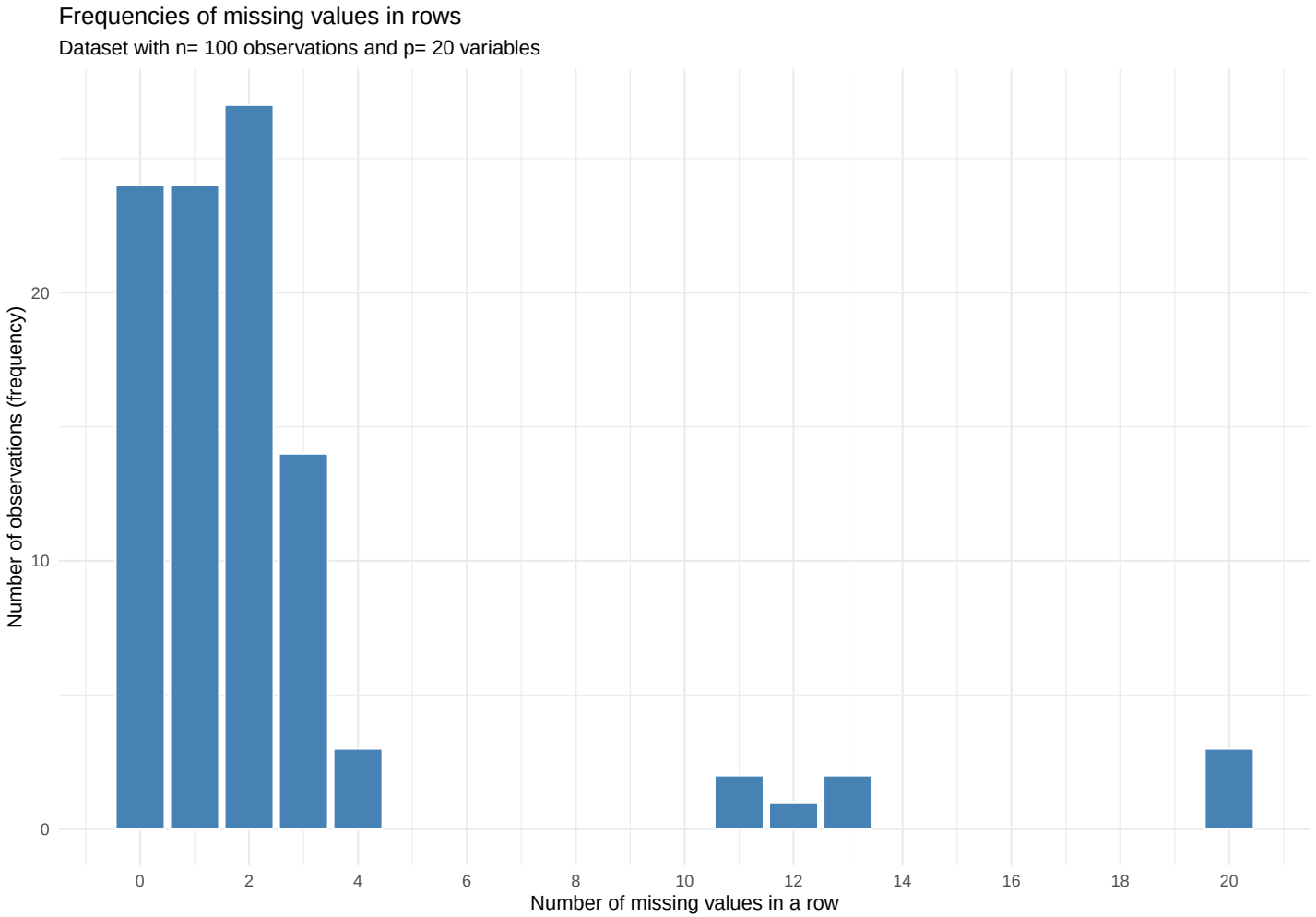
Dealing with missing data

Once we characterized the missing data, **what to do now**?

1. Ignore records with missing values (**delete**?).
2. If categorical, treat missing value as a separate level (or **impute** to an existing one).
3. If numerical, **impute** (fill in) with mean or median, or conditional mean (regression), 1 nearest neighbour imputation, etc.

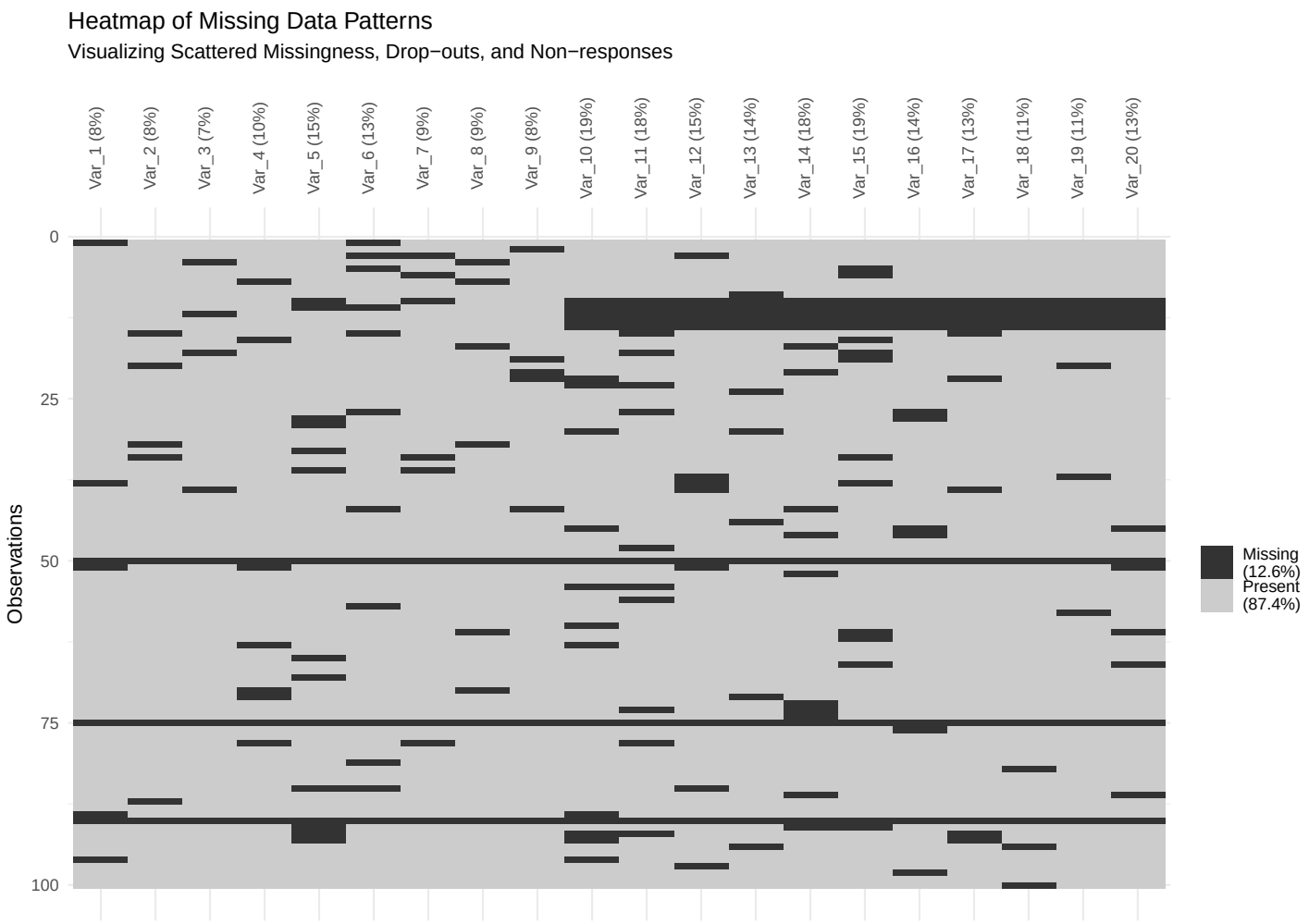
Global analysis of missingness

To get a general picture of the quality of the data we can plot the frequencies of missing values in the whole dataset.



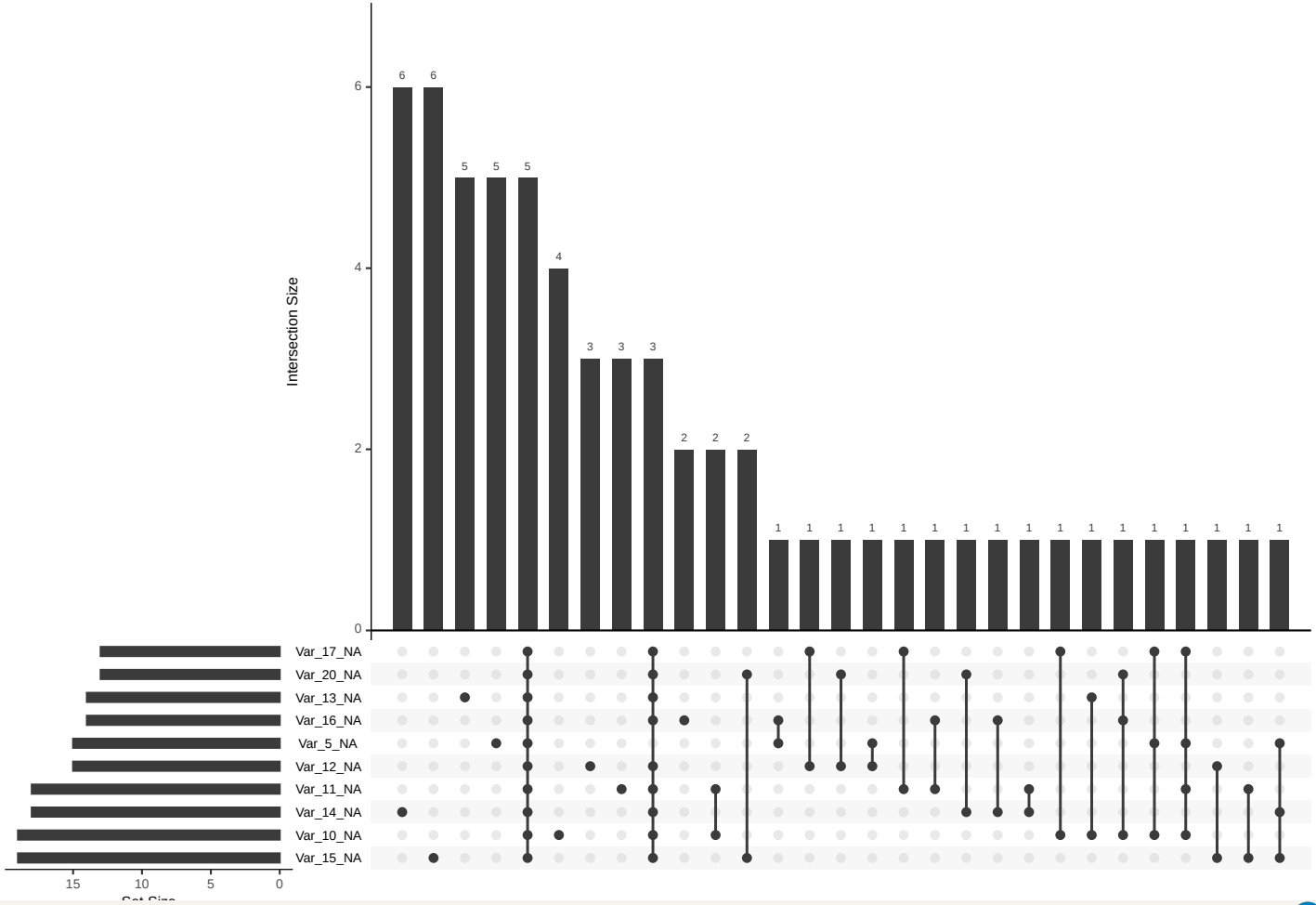
Global analysis of missingness (II)

We can also study the structure of the data quality with a heatmap of the missingness using the base R method `image`‘ or the method `vis_miss` in the library `naniar`.



Global analysis of missingness (III)

While the heatmap shows **where** the missing values are located in the dataset, an upset plot () will give us the intersections between the variables missingness.



Missigness mechanisms

In an statistical analysis is more important understanding **why** data is missing than knowing **what** is missing.

- **MCAR (Missing completely at random):** The missingness is purely a matter of chance. $P(Missing|Data) = P(Missing)$. Removing the row is generally safe.
- **MAR (Missing at Random):** The missingness is related to observed data but not the missing values themselves. $P(Missing|Observed \cap Unobserved) = P(Missing|Observed)$. Imputations method works best.
- **MNAR (Missing Not at Random):** The missingness depends on the unobserved value itself or other unmeasured factors. $P(Missing)$. This introduces a significant **bias** that cannot be fixed by looking at other variables in the dataset.

The `naniar` library provides the `mcar_test` that performs an statistical test to check if the data is in the MCAR category (null hypothesis).

Treatment of missing values

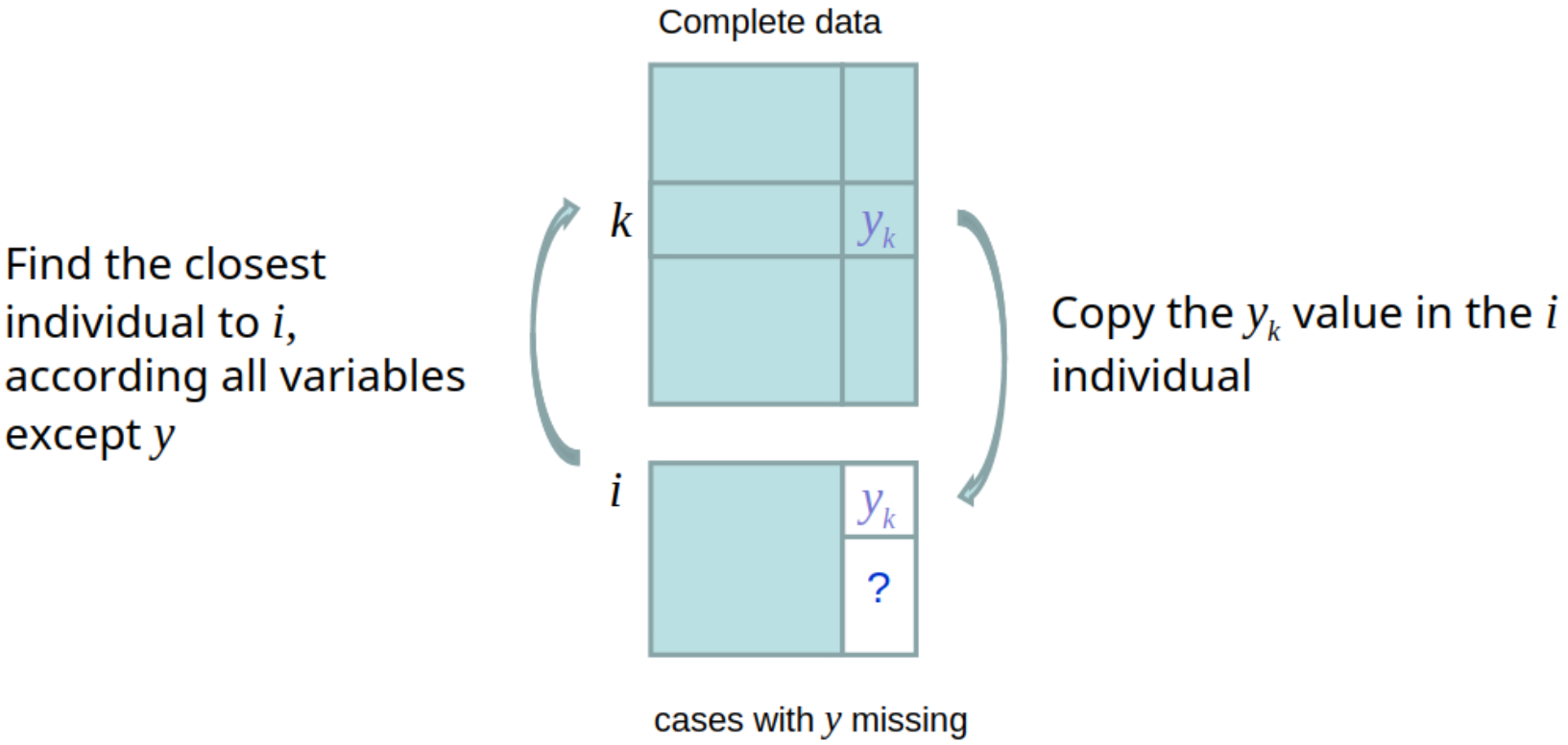
- **Listwise deletion:** Delete individuals with a missing value (information is destroyed).
- **Unconditional mean imputation**: Substitute the missing values of a variable with the mean computed from the non-missing values.
- **Regression imputation:** Perform regression to predict the value of the missing observations. $\mathbb{E}[Y|X]$

Imputing missing values

- **Stochastic imputation:** Simulate actual data by incorporating uncertainty into the random draw.
  $y_{i,j}^{imp} \sim \hat{X}_j + \epsilon$
- **Stochastic regression imputation:** Predict a value for the missing observation using regression and add noise based on the residual variance of the regression.
  $y_{i,j}^{imp} = \hat{y} + N(0, s_{resid}^2)$

k-NN Imputation

The library `DMwR2` offers the `knnImputation` method with **Euclidean** distance. The library `VIM` offers the `kNN` method with **Gower** distance.



Imputation by chained equations

MICE (Multiple Imputation by Chained Equations): Method for complete a dataset by multiple imputation.

1. **Initial fill:** A placeholder value (mean) is used initially for all missing values.
2. **The chain**: Pick a first variable with missing data and use it as the target variable and the rest as predictors in order to perform **stochastic regression**.
3. **Iterate**: Move to the next variable and use it as target for a new **stochastic regression**. Notice that the first variable uses its imputed values from the previous stochastic regression.
4. **Repeat**: Repeat until the values stabilize.

The algorithm repeats the process several times creating m different datasets. A final version of the dataset is obtained by combination of the different datasets.

Imputation by Random Forest

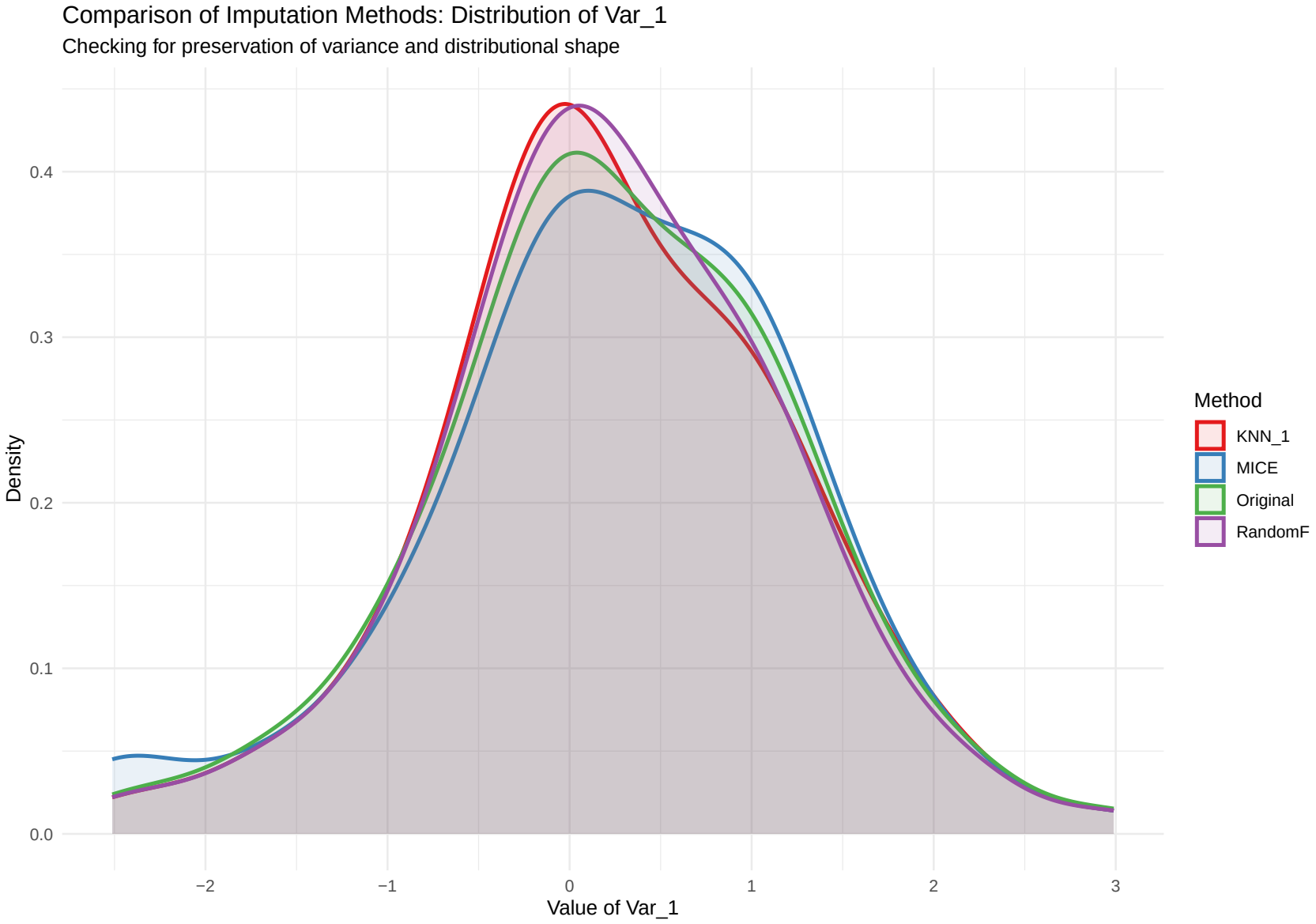
MissForest: A non-parametric algorithm from Machine Learning to fill in missing values.

1. **Initial fill:** Fill all missing values using a simple method (mean).
2. **Sorting:** Variables are sorted based on the amount of missing data.
3. **The chain:** For the first variable with missing data treat it as target and the rest as predictors in order to perform a **Random Forest** regression (or classification if categorical).
4. **Iterate:** Repeat for the next variable using the rest of the variables and the updated variables from previous iterations.
5. **Repeat:** Repeat the loop until the difference between the new and old imputations become negligible.

Imputation methods

- **1-NN:** It only uses values that actually exist in our data so the generating mechanism produces them. Its sensitive to outliers.
- **MICE:** It preserves the distribution (mean and variance). Accounts for statistical uncertainty. Computationally costly.
- **MissForest:** The most accurate method for high-dimensional data as it captures interactions. The **OOB** error can be used to show the model's self-assessed reliability.

Imputation comparison



The missing data summary

Classification	Definition & Context
General Definition	Absence of data for variables in specific observations (represented as NA or NULL).
Missing Values	Individual scattered data points missing across different variables (p) and observations (n).
Drop out	Occurs when a participant stops providing data after a certain point in time.
Non response	Total absence of data for an entire observation across all variables p.
Pros	Can identify systematic collection flaws; allows for simplified analysis if MCAR.
Cons	Reduces statistical power; introduces bias; causes algorithmic failure in PCA/Regression.

Feature engineering

Feature engineering: Process of using domain knowledge to transform raw data into "features" that help methods to perform better. Feature engineering includes:

- **Feature transformation**: Normalization, standardization, codification, discretization, recategorization, etc.
- **Feature selection:** Remove redundant or noisy variables. Some methods do it implicitly.
- **Feature extraction**: Create new features from original ones by aggregating or by some known function.

References

- Little, R. J., & Rubin, D. B. (2019). *Statistical Analysis with Missing Data* (3rd ed.). Wiley.
- Van Buuren, S. (2018). *Flexible Imputation of Missing Data* (2nd ed.). Chapman and Hall/CRC.
- Aluja, T. *Introduction to multivariate analysis.* UPC MVA Course (2020)